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Light Curing of Restorative Materials

RICHARD B. PRICE, FREDERICK A. RUEGGEBERG

The ability to light cure dental resins "on demand" has revolutionized the practice of restorative dentistry and the dental curing light has become an indispensable tool in almost every dental office.[1-3] Despite their ubiquitous presence and routine use, the interaction between curing lights and light-activated dental resins or resin cements is poorly understood.[4] At first glance, it appears that the process of light curing is simple and requires little attention. This misconception is concerning. Although the surface of the resin closest to the light polymerizes readily, giving the appearance that all the resin has been fully "light cured," this has not always occurred in the deeper layers of the restoration. This may result in bulk fracture of the restoration, increased wear, increased release of chemicals, a low bond strength to the tooth, and increased microleakage between the tooth and the resin.[5-15]

In most countries, dental curing lights are classified as medical devices that must pass stringent tests before they are approved for use. This chapter will review the background and discuss the current knowledge of dental curing light and its use in contemporary resin photopolymerization. The International System of Units (SI) terminology that should be used to describe the output from a curing light will also be presented (Table 6.1).

Methacrylate-Based, Free Radical Polymerization

Virtually all resin-based restorative materials used to directly restore teeth belong to the methacrylate family.[1] These methacrylate-based monomers are linked to form polymers by a process called free radical addition polymerization. Once the free radical species are formed, the polymerization process is the same for all types of methacrylate-based restorative materials that are in use today. The only difference is how the free radicals are generated, the rate at which they are generated, and the number of free radicals that are produced. This process consists of the following four steps: activation, initiation, propagation, and termination (Fig. 6.1).

Activation and Initiation

The first step involves activation of a free radical species. This is a compound with an unpaired electron that aggressively seeks another electron with which it can form a covalent bond. There are many types of free radical species, but to become activated they all require an external energy source such as heat, an amine reducing agent (used in "self-curing" or "cold-curing"), or electromagnetic radiation (ultraviolet [UV] light, visible blue light, or microwave energy). In dental resins, the activated, free radical species seek out the electron-rich carbon double bond that is present in methacrylate groups of a monomer molecule. This double bond gives up one electron to the newly initiated radical, and the other electron acts as the free radical agent. This process is called "initiation."

Propagation

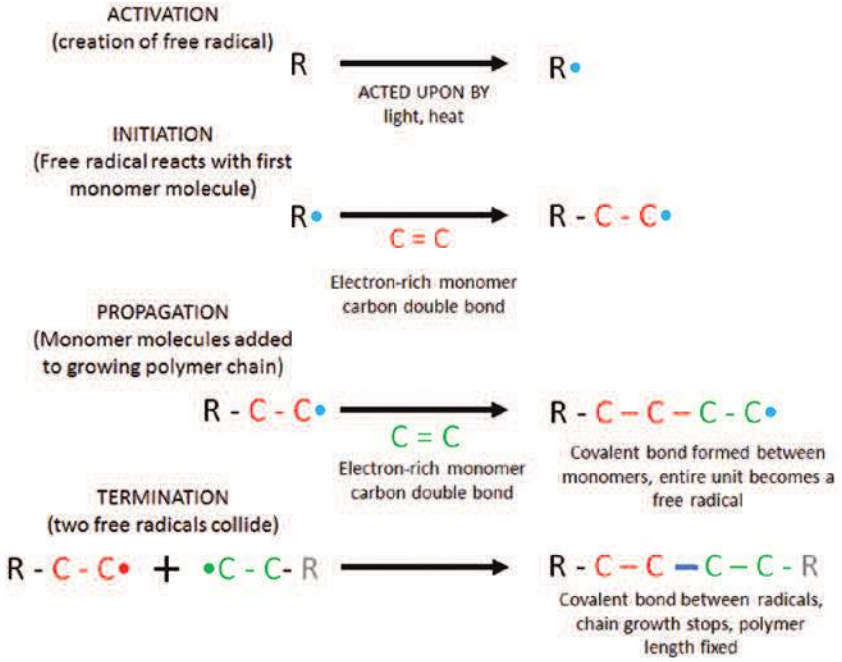
This monomer-radical species then seeks another carbon double bond in an adjacent monomer molecule; it absconds one of the double bond electrons and forms a new covalent bond between the two monomer molecules. This newly attached monomer possesses the unpaired electron, and it brings along the entire polymer chain to find yet more electron-rich carbon double bonds with which it can react. With every additional methacrylate-based monomer that is added to the end, the polymer chain grows longer. This process is called "propagation."

Termination

As the chain length increases, its molecular weight grows, thus decreasing the ability of the polymer chain to move. Also, because the surrounding resin changes from a liquid to a gel, the mobility of the reacting molecules decreases. Further increases in the chain length become diffusion controlled and the inability of the growing polymer chains to readily find and join to additional unreacted methacrylate monomers causes the rate of polymerization to decrease. This phenomenon is termed auto-deceleration. Eventually the polymerized resin becomes a solid, thus further inhibiting linear chain growth. At this point, the probability increases that two free radical ends from two different polymer chains will meet and form a covalent bond. Once this occurs, further increase in the linear chain length becomes impossible. This phase is called "termination." Most direct resin restorative materials used intraorally also contain monomer units that have two or more carbon double bonds that also allow cross-linking between adjacent polymer chains. Cross-linking between chains is highly desirable because polymers that have little cross-linking tend to be weak and flexible, whereas those that are highly cross-linked have greater fracture resistance, are harder, are less flexible, and are more heat resistant.[16] Of note, the use of different curing lights or light exposure modes can result in polymers that have different amounts of cross-linking.[1,16,18] This phenomenon means that solvents, food substances, and enzymatic attack can have different effects on the same dental resin, depending on how it was light cured.[16-18]

TABLE 6.1 Radiometric Terminology Used to Describe the Emission From Dental Curing Lights

Term	Units	Symbol	Notes
Radiant energy	Joule	J	Describes the energy emitted or received
Radiant exposure	Joule per square centimeter	J/cm^2	Describes the energy emitted or received
Radiant energy density	Joule per cubic centimeter	J/cm^3	The volumetric (cm^3) energy density Not to be confused with the energy density term that has been used in dentistry
Radiant power, or Radiant flux	Watt	W or J/s	Radiant energy per unit time
Radiant exitance, or Radiant emittance	Watt per square centimeter	W/cm^2	Radiant power (flux) emitted from a defined area (e.g., the tip of a light-curing unit)
Irradiance (incident irradiance)	Watt per square centimeter	W/cm^2	Radiant power (flux) incident on a known surface area An averaged value over this surface area
Spectral radiant power	Watt per nanometer	W/nm	Radiant power at each wavelength of light in nm
Spectral irradiance	Watt per square centimeter per nanometer	$W/cm^2/nm$	Irradiance per wavelength of light at each nm
Luminous efficacy	Lumens per watt	lm/W	The ratio of luminous flux to power A measure of how efficiently a light source can produce light in terms of the human eye response to light



• **Fig. 6.1** Stages of vinyl, free radical addition polymerization—the mechanism used in polymerization of all methacrylate-based dental resins.

Development of Polymerization Strategies for Dentistry

Self-Curing Direct Restorative Products

The first dental resins were "self-curing" or "cold-cure" acrylics. This reaction combined the same benzoyl peroxide free radical generator that is used in heat-cured resins with an aromatic amine. These components are packaged separately as a "catalyst" and a "base" because when mixed together at room temperature, they react together to form free radicals[19] (Fig. 6.2). The resin component was improved over the years to include dimethacrylate-based monomers and ground glass particles (to make a composite), but the same basic components were used in this self-curing polymerization reaction. Following mixing, the paste was packed into the cavity and held *in situ* for several minutes as the resin polymerized.[19] Additional material could be added to the original increment after it became hard. The material could be used in bulk increments because free radicals are produced throughout the resin as the benzoyl peroxide and the aromatic amine interact. The relatively

slow chemical reaction of these self-curing resin-based restorative materials was a limitation and it was recommended to delay final polishing for 24 hours. Polymerization shrinkage still occurred and was an undesirable outcome.

Light-Cured Polymerization Reactions

A few years later, a faster and less complicated method of free radical polymerization was developed. This process used photons of radiant electromagnetic energy (light) to penetrate a mixture of polymerizable monomers that contained a photoinitiator to generate the free radicals. Only a single component was needed, but it had to be protected from excessive light exposure and inhibitors were included in the system; otherwise the resin would prematurely self-polymerize. After placing this light-curable resin into a tooth, the operator adjusted and contoured the material to achieve the correct shape. The resin restoration was then exposed to light at specific wavelengths and for the specified time to activate the photoinitiator and produce the necessary free radicals. The same sequence of free radical vinyl polymerization then occurred as in the self-curing resins and polymerization shrinkage still occurred.

Radiometric Terminology

To understand the basics of photochemistry and to be able to communicate with others, a background in the proper terminology used in this field is required. Commonly used terms to describe the output from a curing light such as *intensity, power density,* or *energy density* are not SI terms, they are confusing as to what they mean, and ideally they should not be used. The appropriate SI radiometric terms to better describe the output from a curing light are provided in Table 6.1.[3]

When measured at zero distance, such as when the curing light tip is in direct contact with the meter detector or the resin surface, the SI term *radiant exitance* is effectively the same as the SI term *incident irradiance* and is usually described in milliWatts per square centimeter (mW/cm^2). The radiant exitance (*irradiance at 0 mm distance)* is commonly used to describe the output from a curing light; however, this parameter provides limited information about how powerful (Watts) the curing light is, the effect of tip-to-target distance on the irradiance received, the spectral radiant power, or the beam uniformity. For example, if two curing lights deliver the same radiant power, but one has a 7-mm and the other has a 10-mm tip diameter, the smaller diameter tip will be half the area (0.38 cm^2) of the larger one (0.79 cm^2). This decrease in tip area will double the radiant exitance (irradiance) at the tip, even though both lights deliver the same radiant power. Such small tip diameters are often seen in budget curing lights that need only deliver a low radiant power from a small tip to deliver what appears to be an acceptably high irradiance.[20,21] For this reason, both the radiant power (watts) and the irradiance (mW/cm^2) should be reported; otherwise a curing light could deliver a high irradiance, but not be very powerful.

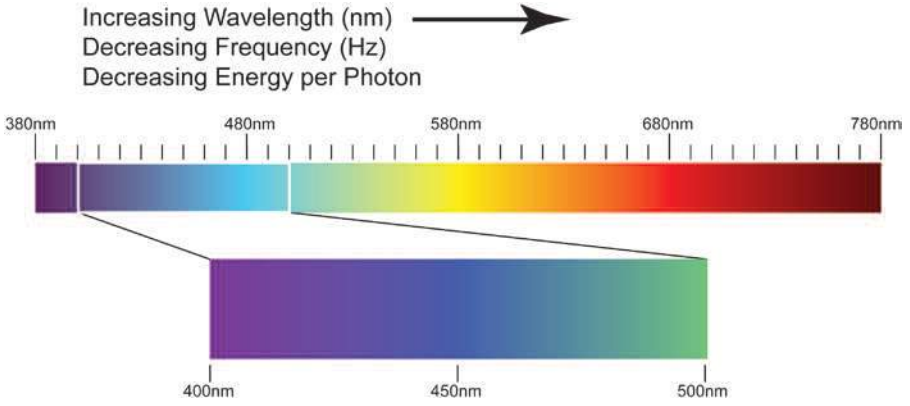


• **Fig. 6.2** Example of a self-curing, paste-paste, resin-based composite. (Courtesy Dental Technologies, Inc.)

Electromagnetic Energy / Electromagnetic Spectrum

The radiant energy perceived by humans as "light" is electromagnetic (EM) radiation. The visible spectrum ranges from red, which is comprised of longer wavelengths of light (700 to 750 nm), through various colors, until reaching violet (comprised of shorter wavelengths between 390 and 400 nm). Fig. 6.3 shows the portion of EM spectrum associated with what humans perceive as visible light, as well as the colors corresponding to those wavelength ranges. EM radiation is delivered in packages of energy called photons and the energy carried by each photon is expressed in the following equation:

$$E = \frac{hc}{\lambda}$$



• **Fig. 6.3** Correlation between electromagnetic energy wavelength (in nanometers) and the human perception of color.

where E is the photon energy, h is Planck's constant, c is the speed of light in vacuum, and λ is the photon's wavelength. The energy carried by each photon changes in direct relation to wavelength λ because h and c are both constants and two photons of the same wavelength will carry the same energy, even if one photon was emitted from a dental curing light and the other from the Sun.

The relationship between the speed of light and wavelength is expressed as the frequency *(f),* where:

$$\frac{c}{\lambda} = f$$

The photon energy equation can be simplified to the Planck-Einstein relation as:

$$E = hf$$

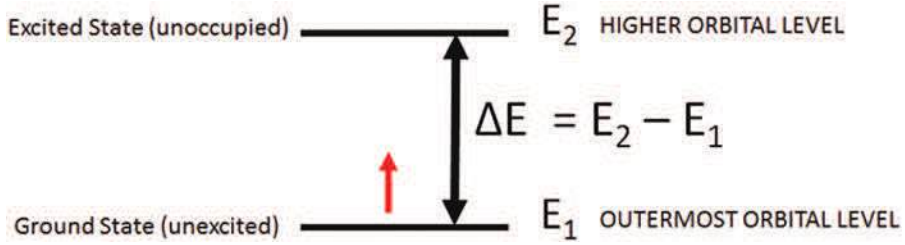
For simplicity, dental terminology utilizes wavelength (in nanometers) rather than the frequency (Hz) to describe the properties of light, although both are directly related. Longer wavelength electromagnetic energy (lower frequency) than what the human eye can perceive as the color red is in the region of the EM spectrum known as infrared (IR) radiation. Photonic energy delivered in this region interacts with molecules of matter and increases their vibrational amplitudes. This interaction increases the probability that two molecules will collide with each other, producing friction and ultimately heat. In contrast, photons of shorter wavelengths (higher frequency) in the ultraviolet (UV) range contain more energy than those in the longer wavelength range (red color). Photons in the UV range are considered to be ionizing because they deliver sufficient energy to totally remove an electron from the outer shell of an atom, creating a negatively charged species: an ion.

Light Curing of Restorative Resins

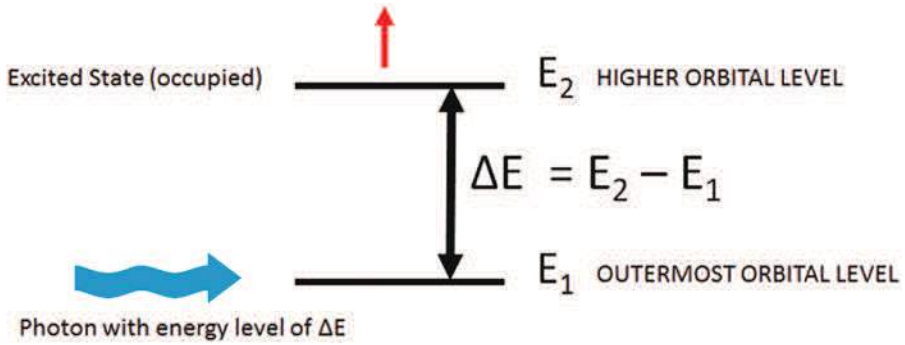
The Photoinitiator Molecule and Free Radical Generation

The interaction between EM radiation and matter is used in the various curing mechanisms to create the free radicals that are required to polymerize dental resins. For a photochemical reaction to take place, the first law of photochemistry, the Grotthuss-Draper law, states that light must first be absorbed by the chemical substance. The second law of photochemistry, the Stark-Einstein law, states that, for each photon of light absorbed, only one molecule within the chemical system can be activated. These two laws, together with the Planck-Einstein relationship, explain why two curing lights that deliver the same radiant power, but at different wavelengths, may produce different results.

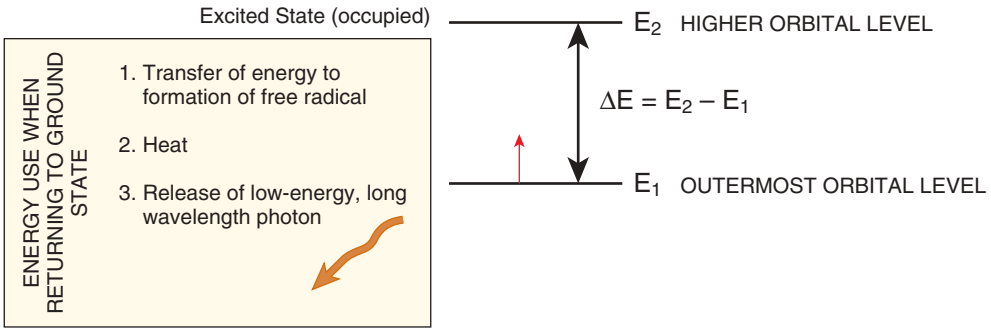
The photoinitiator molecules used in light-curable resin materials contain specific types of bonds that are only capable of absorbing EM radiation within certain wavelength ranges. The interactions of a photon with electrons present in the outermost orbital level of a photoinitiator atom are diagrammatically described in Fig. 6.4. When the photoinitiator atom is struck by a photon at the appropriate energy level (wavelength), an electron in the outermost orbital level absorbs the energy, thus quenching the photon (Fig. 6.5). This electron is then raised to the next higher orbital (quantum) level (E_2) and the excited atom now contains potential energy that can be transferred (Fig. 6.6) because in this excited state the electron is unstable and rapidly decays its ground state level. This causes chemical reactions to occur that create free radicals that initiate the free radical polymerization process. If no reaction occurs, the electron falls back to its ground state and returns the absorbed



• **Fig. 6.4** Prior to absorbing a photon, an electron exists in an outer orbital shell at a specific quantum energy level—its "ground state" *(red arrow).* A discrete amount of energy exists between this orbital level (E_1) and the higher level (E_2, designated as ΔE).



• **Fig. 6.5** When struck by a photon of light at the correct wavelength (frequency) and energy level (ΔE), the electron in the lower orbital shell absorbs this energy and is raised to a higher orbital level *(red arrow)* that is dependent on the energy difference.

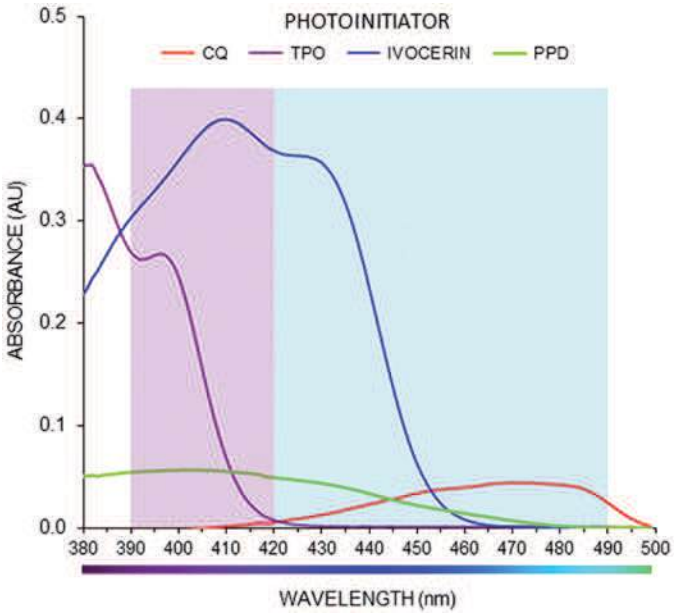


• **Fig. 6.6** Upon its return to the original ground state orbital level, the stored energy is released from the electron.

energy as heat as well as a longer wavelength photon. The ability of a photoinitiator to interact with electromagnetic energy is dependent on the frequency (wavelength) of the photon and the very specific electron configuration of the photoinitiator molecule. If these two needs do not match, then no radicals are formed. Thus in accordance with the first and second laws of photochemistry the critical parameter for effective photocuring is the *number of photons of the correct wavelength that are absorbed by the photoinitiator system(s)* within the resin composite.

Generally, two types of photoinitiators are used in dentistry: Type I photoinitiators have a high quantum yield, meaning they require fewer photons to generate a free radical than do Type II initiators.[22-24] Type II photoinitiators (e.g., camphorquinone [CQ] and 1-phenyl-1,2-propanedione [PPD]) also require an additional secondary electron transfer agent (this is typically an amine electron accepting agent) to generate a free radical.[22] Thus Type II photoinitiators are slower and less efficient than Type I initiators (e.g., Lucirin TPO, and derivatives of dibenzoyl germanium such as Ivocerin). These Type I initiators do not require additional co-initiators and break down directly into one or more free radicals when they absorb light at the correct wavelengths.

Fig. 6.7 indicates the spectral absorption profiles of four photoinitiators present at equivalent molar concentrations, and the relative ability of each photoinitiator to absorb radiant energy at each wavelength. It can be seen that CQ is best activated by light in the blue region close to 470 nm, whereas Lucirin TPO absorbs light only within the violet region below 420 nm, with a maximum absorption near 385 nm.[23,24] Initiators such as Ivocerin and PPD are reactive to a broader-spectrum of light because they are activated by EM radiation, both the violet and blue color ranges. For example, Ivocerin is most reactive near 410 nm but is still very sensitive to wavelengths of light between 400 and 430 nm.[24]



• **Fig. 6.7** Spectral absorption profiles of the most commonly used photoinitiators in dentistry, when all are at equivalent molar concentration. As a reference, the bandwidths associated with "violet light" (390 to 420 nm) and that of "blue light" (420 to 490 nm) are also provided to present the potential for interaction between these photoinitiators and the output from a dental curing light: For example, some LED units deliver no light below 420 nm.

Ultraviolet vs. Blue Light

Initial attempts to light cure directly placed resin-based restorative materials used UV light at wavelengths shorter than 400 nm. Although the polymerization reaction was under the direct control of the clinician, the inability of these short UV wavelengths to penetrate deeply into the resin composite meant that the incremental layers of resin could be at most 1 mm thick. This limitation increased the chairside treatment time required to restore a tooth and did not save much time when compared to using a self-curing product.[25] Although some of the UV-cured materials were successful when placed by excellent clinicians,[26] concerns that the UV light could cause cataract formation or selective changes in the oral bioflora meant that the use of UV light was discontinued. Instead a different light source and photoinitiator, camphorquinone (CQ), was used. CQ is very reactive both to EM radiation shorter than 320 nm and to blue light, with an absorption peak near 470 nm. The initial curing lights developed to activate CQ used a quartz-tungsten-halogen (QTH) filament-based light source, similar to a projector bulb. This QTH light source emitted a broad spectrum of radiant energy, from near UV through visible light and well into the infrared region. Aggressive optical filtering was used to eliminate IR wavelengths that might generate excessive heat, or UV wavelengths that deliver ionizing EM radiation. Thus the curing light delivered only blue light between approximately 400 and 500 nm. This blue light can penetrate deeper into the resin than UV or violet light, such that some "blue light–cured" restorations can now be adequately light cured in increments that are up to 5-mm thick. This enhancement reduced chairside treatment time and provided a "set-on-command" restorative product. However, polymerization shrinkage still occurred and remains a concern.

Factors Affecting the Ability to Polymerize a Resin-Based Composite

Depth of Cure

Within a composite, the degree of conversion (the extent to which resin monomer has been converted into polymer) and the depth to which the resin is cured are related to the radiant exposure (the number of photons) received by the resin (J/cm^2) (Table 6.1). Where insufficient light is delivered, the resin is inadequately polymerized and the boundary between what is considered the cured and uncured resin is called the *depth of cure*. Under optimum light curing conditions, this point is reached at a depth of 1.5 to 2 mm from the surface for a conventional resin composite and between 4 and 6 mm for a bulk-fill resin composite. In cases of poor access to the restoration, or when using more opaque shades, the depth of cure will be less.

Exposure Time

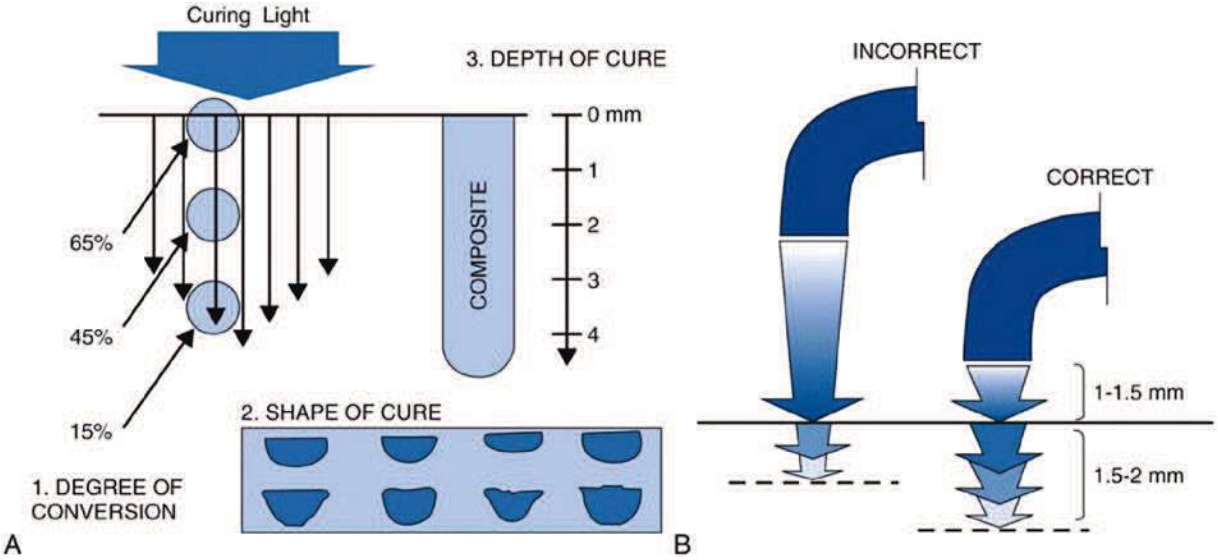
Under optimal conditions using a curing light that delivers 1000 mW/cm^2 from an 8- to 10-mm-diameter tip, many resin composites require a 20-second exposure time typically (delivering 20 J/cm^2) to adequately cure a 2-mm-thick increment of resin. Increasing the concentration of photoinitiator in the resin composite system, reducing the inhibitor concentration, matching the refractive indices of the resin and the filler, and, to a limited extent, increasing the irradiance can shorten this time. However, true reciprocity between the duration of exposure and the irradiance does not exist and short exposure times are not recommended.[14,27-31] If the manufacturer recommends delivering 500 mW/cm^2 for 40 seconds (delivering 20 J/cm^2), this does not mean that the same extent or type of resin polymerization would occur if 4000 mW/cm^2 was delivered for 5 seconds, even though both resins would receive an identical radiant exposure, 20 J/cm^2. However, it is possible to somewhat compensate for a lower irradiance (e.g., if a weaker light is used, or if the distance between the light tip and the top surface is increased) by following the instructions for use and increasing the exposure time.

Effect of Thickness of the Restorative Material

The number of photons of light that reach the depths of the resin depend on many factors, which include the thickness of the resin increment, the wavelengths of the light delivered, the refractive indices of the resin and filler components, the filler content and size, and the overall opacity of the material.[32,33] As the resin composite thickness increases, exponentially fewer and fewer photons reach to the bottom, thus potentially reducing the degree of conversion of the resin at the bottom.[34-37] Even small increases in resin thickness can have a large effect on the amount of transmitted light and resin polymerization such that doubling the exposure time cannot compensate for a doubling of the increment thickness (Fig. 6.8).

Interaction Between Wavelength of Light and Filler Particles

The influence of the Rayleigh scattering of light somewhat helps to explain why the shorter "violet" wavelengths of light do not penetrate as deeply as do longer wavelengths of "blue" light, into dental resins,[1,2] and it has been reported that light scattering is greater when the filler particle size is equal to or smaller than the wavelength of light (i.e., less than 460 nm or 0.46 microns) (Fig. 6.9).[32,38] Although the Beer-Lambert law plays a role in the amount of light penetrating through the resin, this law can only partly predict the amount of light transmitted through the resin material because there is a complex interaction between the filler particles scattering the light, the colorants that absorb the light, the photoinitiator(s) that use the light, and the differences between the refractive indices of the resin and the fillers.[38,39] Manufacturers will often adjust the type and concentration of photoinitiators, inhibitors, and fillers for different shades of the same product to compensate for this decreased light penetration and to optimize

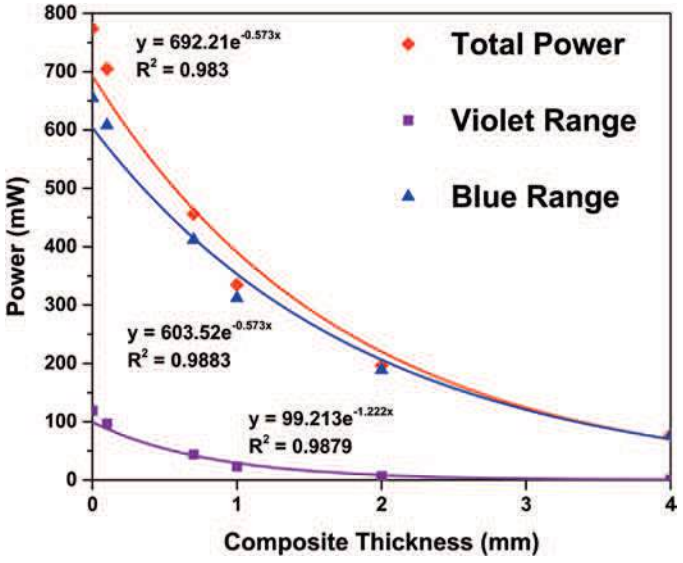


• **Fig. 6.8** Influence of irradiance on the polymerization zone in a resin composite. A, Varying the irradiance with width and depth affects the degree of monomer conversion, the shape of the cured material, and its depth. B, The proximity of the curing light tip to the restoration surface affects the number of photons reaching the top surface and thus the monomer conversion throughout the restoration.

the exposure time. Consequently, the resin manufacturer instructions should be followed because an opaque, but white, shade of composite may require a longer exposure time than a yellow, but more transparent, shade.

Matching Resin and Filler Refractive Indices

The closer the refractive indices of the resin and the filler are matched, the better will be the light transmission through the resin composite and the better the depth of cure.[32,38,39] If they are perfectly matched, the filler particles will optically disappear, but the increased translucency will mean that the resin composite will appear gray in the mouth. In the unpolymerized state, the refractive index of the resin is often lower than that of the filler. As polymerization occurs, the refractive index of the resin component increases. Consequently, most, but not all, resin composites become more translucent after light curing. For this reason, shade selection of composite should not be performed using uncured resin composite paste, but instead using the light-cured product.



• **Fig. 6.9** Differences in the loss of "blue" (>420 nm) and "violet" (<420 nm) wavelengths of light when passing through a commercial resin-based composite. The light in both regions decreases exponentially with increasing composite thickness, and almost no violet light penetrates beyond 3.5 mm of resin composite.

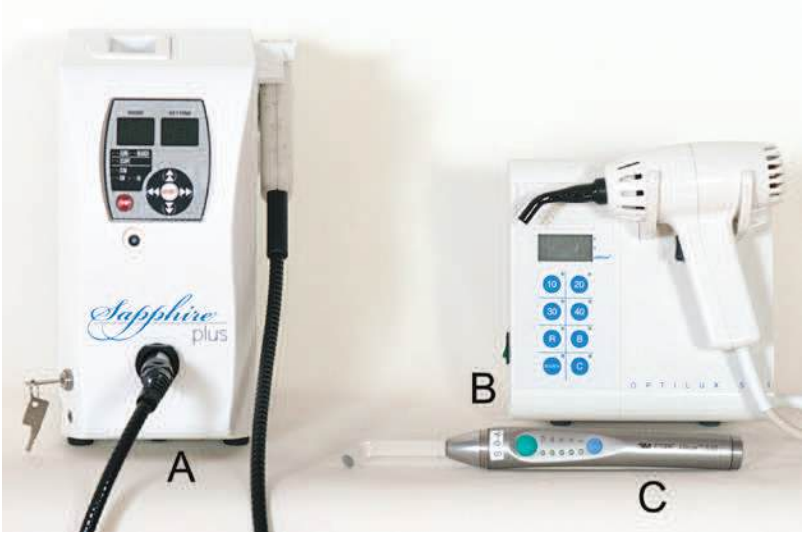
Development of Dental Curing Lights

There are four types of blue light sources that have been used to activate the photoinitiators in dental resins: quartz-tungsten-halogen (QTH), plasma-arc (PAC), argon-ion laser, and the light-emitting diode (LED). These sources produce light in different ways, and guidelines for the selection and use of curing lights have been proposed.[1-3,40-42] Examples of PAC, QTH, and LED curing lights are shown in Fig. 6.10.

Quartz-Tungsten-Halogen Lights

The first visible curing lights used QTH low-voltage 35- to 150-watt projector bulbs (one water-cooled unit used a 300-watt bulb).[1,42] The QTH source consists of a tungsten filament that is surrounded by a clear, crystalline quartz bulb containing a chlorine-based halogen gas. When electrical current passes through the filament, the tungsten wire becomes incandescent and atoms are vaporized from the filament surface. This vaporization releases a large amount of electromagnetic energy, most of which falls into the infrared region. When the current is turned off, the filament cools and the vaporized tungsten atoms are redeposited from the halogen gas back onto the filament surface in a process termed "the halogen cycle."

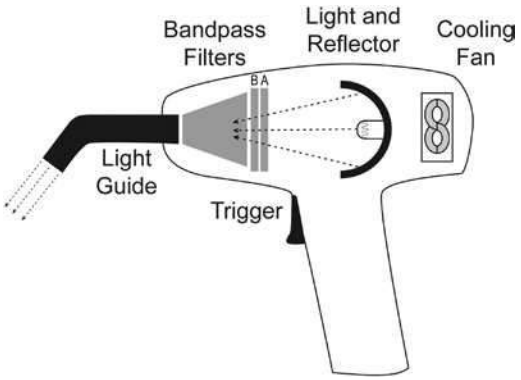
To prevent the QTH light from delivering significant amounts of unwanted EM radiation to the resin and tooth, the optical radiation from the QTH bulb must be filtered. The light is first directed toward a silverized parabolic reflector behind the bulb. The reflector acts as dichroic filter that allows some of the longer wavelengths of infrared light to pass through and not be reflected forward. The shorter wavelengths of visible light are reflected forward toward an infrared bandpass filter (see Fig. 6.10A) to block the remaining infrared wavelengths. The light then passes through a visible light bandpass filter (see Fig. 6.10B). Thus only light at



• **Fig. 6.10** Examples of a PAC (A), QTH (B), and LED (C) curing light.

wavelengths between 400 and 500 nm reaches the proximal surface of a multistranded, bundled, glass fiberoptic light guide, where it is transmitted through hundreds of small optical fibers to the tip end. QTH curing lights are very inefficient; it has been estimated that as much as 70% of the input power to the QTH bulb is converted to heat, with only 10% producing visible light, and only 0.5% to 2% of the total energy input is emitted as useful blue light.[42,43]

These QTH units were developed using a handheld, gun-style form factor, with the QTH bulb located within the body of the unit (Fig. 6.11). Although an internal fan dissipates some of the unwanted heat away from the bulb, filters, and reflector, the reflector surface and QTH bulb still become very hot when the light is on, and then cool down between uses. Vapors from bonding systems, solvents, cleaning agents, or even moisture in the operatory air can become deposited onto the reflector, thus dulling or clouding its surface. If this occurs, the reflector or the lamp unit should ideally be replaced, but they can be carefully cleaned with alcohol or methyl ethyl ketone solvents on cotton swabs. Under ideal conditions, the QTH bulb should last for about 50 hours,[1,2] but the lifespan will be shortened if the operator immediately turns the power supply off after use to stop the noise from the fan. This action prevents the halogen cycle from occurring because if the bulb does not cool down at a controlled rate, the vaporized tungsten atoms are not properly deposited back onto the filament surface. This permanently darkens the inner surface of the quartz glass, thins the tungsten filament, and shortens the lifespan of the bulb. All these changes can occur without any outward sign to the dentist that the output from the curing light is decreasing.



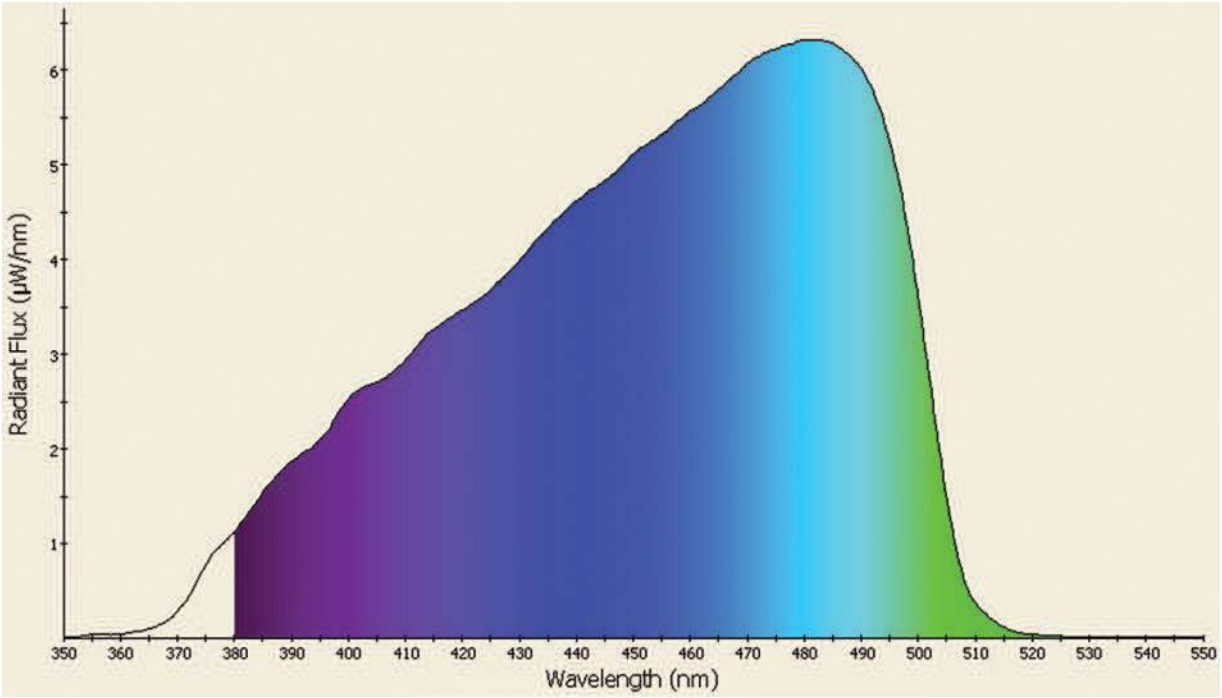
• **Fig. 6.11** Internal components of a quartz-tungsten-halogen (QTH) curing light, using a gun-style form factor and pistol-like trigger activation. A, Infrared light filter; B, band pass filter allowing only light between 400 and 500 nm to pass through.

The emission spectrum from a typical QTH light source developed is shown in Fig. 6.12.

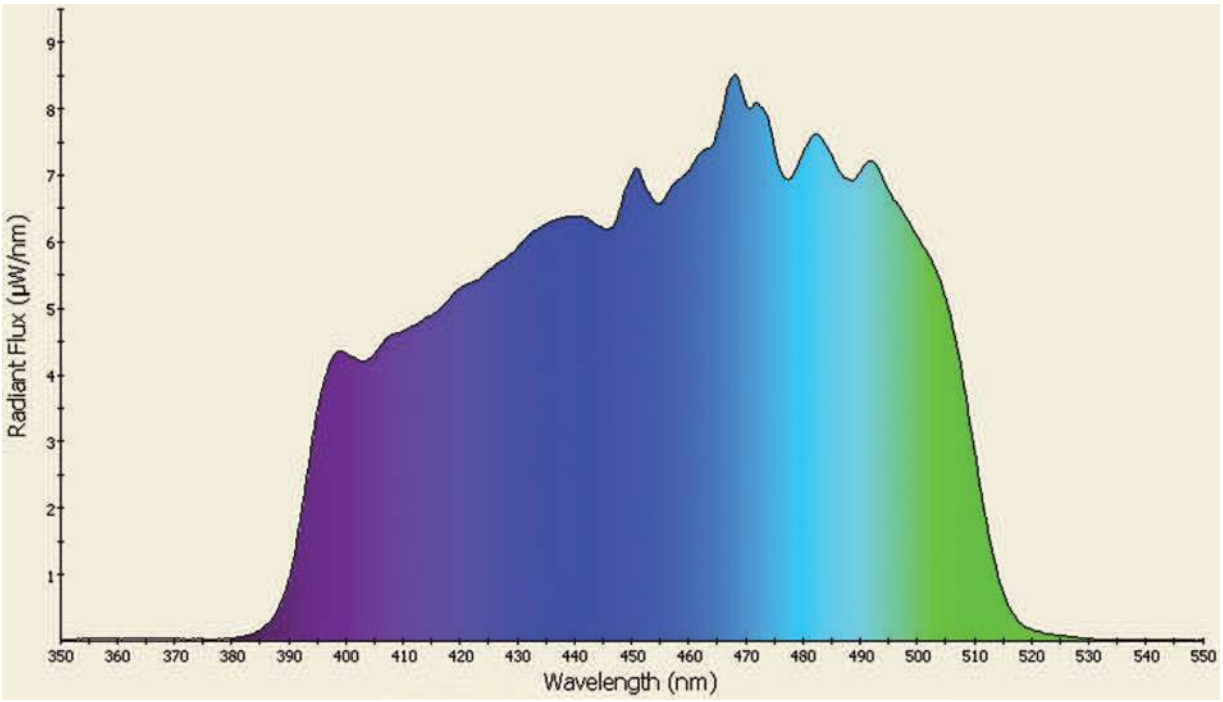
Although QTH lights deliver a broad emission spectrum, most of these units were mains powered. The cooling fan was noisy, and the units delivered a relatively low radiant power and low irradiance. Consequently, they required an exposure duration between 30 and 60 seconds to adequately polymerize a 2-mm-thick increment of resin composite.

Plasma-Arc Lights

In an attempt to reduce light exposure times, high-power PAC lights were introduced. These lights were promoted to deliver a very high irradiance and the initial units claimed that exposure times between 3 and 5 seconds could successfully light cure a 2-mm-thick increment of resin composite. Instead of a filament, the PAC light source has two tungsten electrodes that are surrounded by xenon gas. When a high voltage is applied, a spark is formed that ionizes the gas.[1] This spark then acts as both a light emitter and an electrically conductive gaseous medium (a plasma) to maintain the spark. The gap between the electrodes is parallel to the long axis of a parabolic-shaped reflector that directs the emitted



• **Fig. 6.12** Emission spectrum from a typical QTH curing light (Demetron 400, Kerr Dental).



• **Fig. 6.13** Example of the broad emission spectrum from a PAC curing light (ARC Light II, Air Techniques).

light forward and through a sapphire window. Extreme optical filtering is used to prevent both the emission of unwanted ionizing radiation and infrared wavelengths of light that would overheat the tooth.

The emission spectrum of a typical broad-spectrum PAC curing light is shown in Fig. 6.13. Most PAC lights deliver a broad emission spectrum from 390 to 510 nm, a high radiant power, and a high irradiance. However, an early brand of PAC light used multiple light tips to provide different light outputs and outcomes: a bleaching tip (full spectral output), a 430-nm tip for photocuring materials containing the alternative initiators, and a 470-nm tip for light curing CQ-containing materials. This use of specific light tips to deliver different emission spectra for particular situations led to confusion and clinicians were unsure which light guide they should use to cure the specific brand of resin they were placing. Initially, this PAC light could only run for 3 s, but later generations of PAC lights used more sophisticated electronics to control the voltage to the bulb and could provide a high light output for 60 minutes or more.[1] However, PAC lights are expensive, noisy, large, not portable, cannot be battery operated, and have become less popular with the introduction of high-output LED curing lights. In addition, the xenon bulb is expensive to replace when it fails.

Argon-Ion Lasers

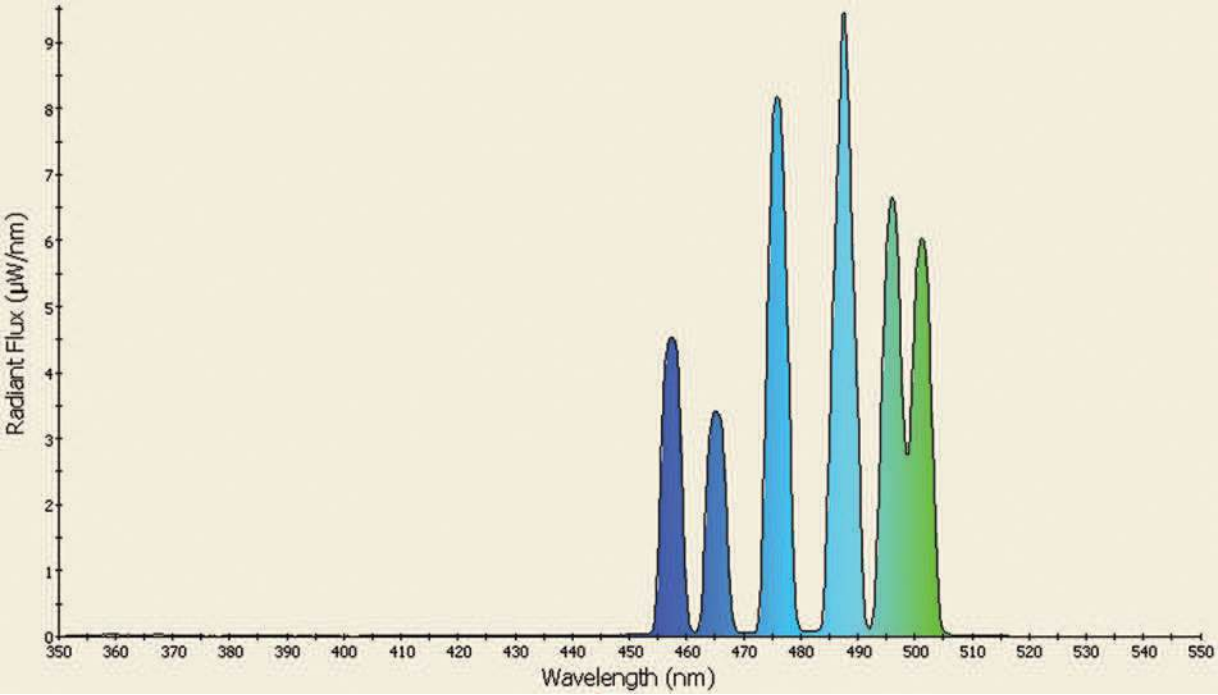
The term "laser" is an acronym for **l**ight **a**mplification by **s**timulated **e**mission of **r**adiation. Argon-ion laser curing lights also claimed short exposure times of 10 seconds or less compared with 40 to 60 seconds using a conventional QTH source.[1] An example of a laser curing light is shown in Fig. 6.14.

These curing lights work by delivering electrical energy to specific atoms present within the light emitter. These electrons then become excited and move from a low-energy orbit to a high-energy orbit around the atom's nucleus. When these excited electrons return



• **Fig. 6.14** Laser curing light (AccuCure 3000, LaserMed, Inc).

to their normal or "ground" state, they emit photons of light. These photons are all at the same wavelength, which is determined by the amount of energy released when the electron drops back to a lower orbit. The internally generated laser light is directed to an emitting tip through a thin fiberoptic cable, where the size of the light beam is expanded before it is directed toward the target. The argon-ion laser does not produce a broad spectrum of light



• **Fig. 6.15** Emission spectrum from an argon-ion laser curing light (AccuCure 3000, LaserMed, Inc.).

but instead produces several intense and well-defined emission peaks in the blue spectral region. Thus argon-ion lasers were considered to be a viable option for a high irradiance curing light that could rapidly cure dental resins. Fig. 6.15 shows the narrow emission spectrum from a typical argon-ion laser curing light. With the introduction of LED curing lights, lasers have become less popular because they are expensive, not portable, have a narrow emission spectrum, and cannot be battery operated.

"Turbo" Light Guides

To reclaim some of the market from the PAC and laser curing lights, two features were added to QTH curing lights to boost their irradiance and thus attempt to reduce exposure times. The first was the so-called "turbo-tip." At that time, conventional glass-fibered light guides came in a variety of diameters, but the proximal entrance aperture and distal exit diameters of the light guide were the same size. To increase the irradiance (the number of photons emitted over the output area), the entrance to the turbo light guide was made larger than the exit diameter.[44,45] This enhancement was accomplished by using hundreds of glass fibers that were larger in diameter at the proximal end (nearest to the light source) than the opposite (distal) end, nearest to the exit tip. Thus many photons could enter the large entrance to the light guide, and the same number (assuming no loss) of photons were then emitted over a smaller area at the exit tip. This arrangement delivered the same radiant power (watts) as would come from a conventional light guide placed on the same device, but over a smaller tip area. This modification increased the irradiance (mW/cm^2) to a value similar to that from PAC lights. Some QTH lights also included a special boost mode, where the voltage to the bulb was raised for a short time, thus increasing light output. By combining the increased irradiance from the small diameter turbo tip with the increased emission from a QTH bulb that was powered above its maximum voltage rating, a higher output was obtained from the QTH unit. Unfortunately, this procedure shortened the life expectancy of the QTH bulb and the optical design of the turbo tip increased the irradiance for only a few millimeters from the tip end. The irradiance from these turbo tips then declined rapidly as the distance from the tip increased and the light beam dispersed.[44,45] This dispersion adversely affected the extent of resin polymerization when the light tip was more than 5 mm away from the resin[46] and alerted researchers to the fact that it is not the light output at the light tip that matters, but it is what is received by the resin that is important.[45]

Stress Development During Polymerization

Fast polymerization of dental resin composites is thought to adversely affect the mechanical properties of the polymer network.[1,47,48] This phenomenon occurs because, when the reaction rate is very fast, the liquid monomer is quickly converted to a solid, and the polymerization reaction rapidly becomes diffusion limited.[49] Thus, in some contemporary dental resins, rapid photopolymerization produces undesirably short polymer chain lengths because there is simply insufficient time to form many long chains before resin solidification is reached.[47] In addition, the formation of the monomer-to-monomer bonds also causes the resin to shrink, thus decreasing the overall net volume of the system. As long as the system is in a liquid state, it can physically deform and no stress develops; however, beyond the gel point, the resin becomes a solid and further polymerization shrinkage creates strain both within the resin network and at the interfaces between the tooth and the resin. When the polymerization reaction occurs rapidly rather than slowly, the gel point is reached sooner, the resin becomes hard sooner, and these outcomes may result in increased stress,[50] bond failure, and increased gap formation between the tooth and restorative material.[51,52] Ultimately, these consequences can lead to premature restoration failure, cusp fracture, or increased tooth sensitivity.

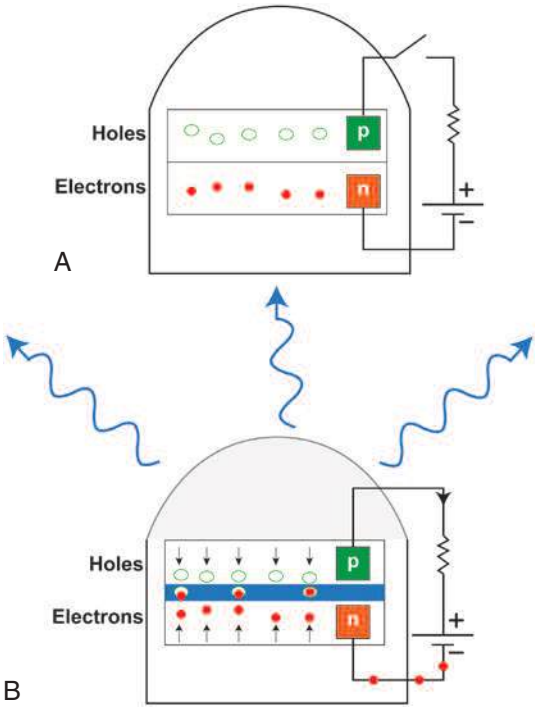
Soft-Start Exposures

With dentists and manufacturers trying to achieve ever shorter exposure times by using lights that delivered a high irradiance, concerns arose about the potential for damaging increases in intrapulpal temperature as well as the potential for generating excessive internal stresses when using fast-shrinking, high-modulus resin composites.[1] In an attempt to decrease these effects, a variety of different curing modes were developed. An example is the "soft-start" curing mode, where a low irradiance value is initially applied to the restoration surface for a short time, and then either immediately, or over a period of time, the light output increases to its full operating level for the remainder of the exposure.[52,53,54] Theoretically this technique reduces the rate at which the photocurable material reacts, produces a more desirable polymer structure, allows the partially cured resin composite to deform at its unbonded surface, reduces the stresses developing at the bonded interface, and reduces the polymerization contraction stress.[50,55-58] Another type of soft-start method (the "pulse delay") delivered an initial low irradiance, short-duration exposure. After waiting some 3 to 5 minutes, the final exposure was provided at the full light output.[52,53] This technique may work *in vitro* for certain resin composites polymerized by specific curing lights.[52,57,59] However, some evidence suggests that soft-start polymerization can result in polymers that have different amounts of cross-linking and that solvents, food substances, and enzymatic attack can have different effects on the same dental resin, depending on how it was light cured.[16-18] It is thought that any potential correlation between enhanced marginal adaptation, reduced shrinkage, and lowered polymerization contraction stress using soft-start polymerization is probably due to the resin receiving less total energy and consequently reaching a lower degree of conversion.[1,60] None of these attempts to reduce the polymerization shrinkage rate and reduce stress development have yet been found to provide better clinical performance compared to continuous light exposure,[1,61] possibly because of the irradiance levels used or the different exposure times used by commercial curing lights to ramp up to full power. Some soft-start exposures in commercial curing lights can be as short as 2 s,[62] whereas in the encouraging *in vitro* tests there was up to a 5-minute delay between the initial light and the final light exposure.[52]

Light-Emitting Diode Technology

The next innovation in dental photocuring came in the 1990s, with the development of blue light-emitting diodes, or LEDs.[63,64] These LED curing lights have many advantages over their predecessors and they dominate the market.[2,42] The LED emitters are solid state, lightweight, battery driven, and more efficient; do not require any bandpass filters; and have emitting sources that can provide an extremely long working life when compared to any filament or spark-based light source.[1,2,63,64] LEDs are also very efficient at converting electrical energy to electromagnetic radiation. Their luminous efficacy (measured in lumens per watt) was initially at least twice that of a QTH bulb, but, with continued improvements, some are now 10 times more efficient.[2]

LEDs are semiconductor light sources that have been doped with impurities to create a p-n junction. Light is produced by interposing the electron-excessive (n) and electron-depleted (p) surfaces of two different semiconductor materials in an electrical circuit.[2] When a current is applied, this p-n junction emits light by a process called electroluminescence. The semiconductor substrate



• **Fig. 6.16** Light is emitted from the p-n junction of a light-emitting diode (LED) when electrons from the n substrate pass into the holes from the p substrate, as electrical current flows through the circuit.

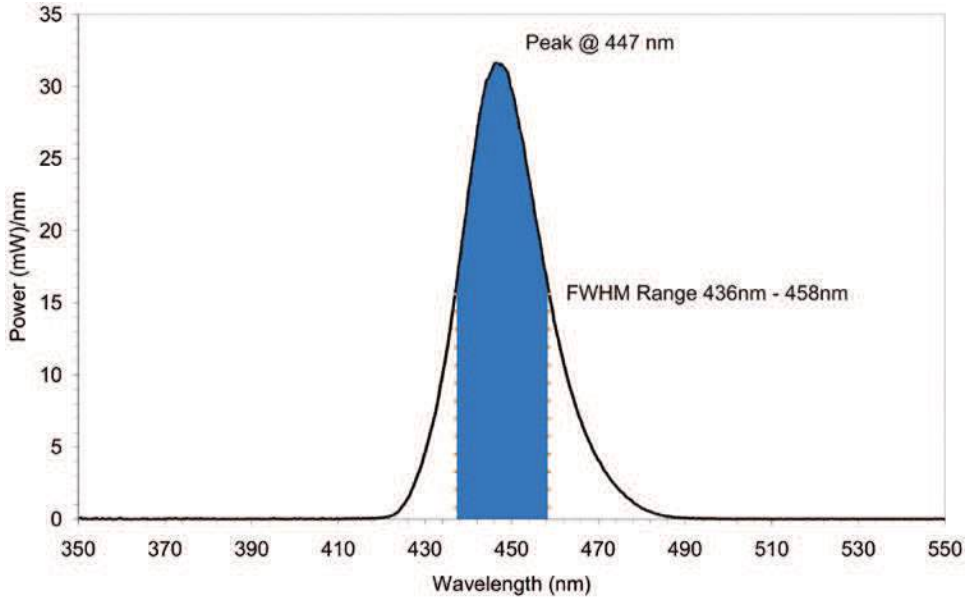
on the n side, or cathode, has an excess of electrons and will rise in potential when supplied with sufficient electrical energy. Electrons from the n substrate then pass through and "fall" into "holes" in the electron-deficient p side, or anode. When an electron meets a hole, it "falls" into a lower energy level and releases energy in the form of a photon of light, as illustrated in Fig. 6.16.

The color of the light emitted corresponds to the energy of the photon that is in turn determined by the composition of the two semiconductors and their resulting "band gap" potential.[2] The semiconductor material used in blue LEDs is composed of a mix of gallium nitride (GaN) and indium nitride (InN). This combination produces light in the blue spectral range that has a relatively narrow bandwidth with a full-width half maximum (FWHM) range of only about 20 nm (Fig. 6.17).

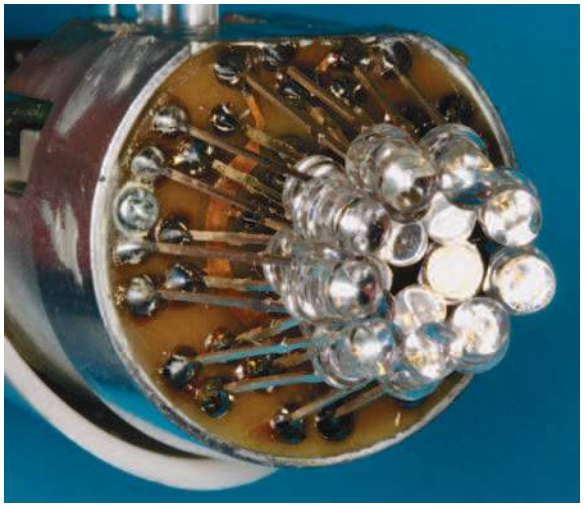
First-Generation LED Lights

The early versions of these energy-efficient, gallium nitride, blue light–emitting LED curing lights emitted blue light from an array of individual, but similar, low-power LED casings (cans). Although these first-generation blue LED curing lights (Fig. 6.18) delivered a low radiant power output, they generated great interest. They were portable, battery operated, and lightweight, and the blue wavelength output from these early LED chips was in the range of maximum absorption of CQ (470 nm).[1] This made these blue LED lights very efficient at producing the free radicals required to photocure CQ-containing dental resins.

When the first-generation LED lights were marketed, they were promoted as "cool" lights that produced less temperature rise in the pulp than did QTH lights,[65-67] and it was initially thought that all LED units would generate little heat—something that was becoming a concern. However, in retrospect, it is now recognized that the claims of low heat generation from the early LED lights



• **Fig. 6.17** Emission spectrum of a typical single-peak LED curing light showing a typical full-width half maximum (FWHM) bandwidth of about 22 nm of blue light.



• **Fig. 6.18** Example of the internal configuration of a first-generation LED curing light that contained multiple, individual low-power LED "cans."

were due to their low radiant power output rather than to any property of the light they emitted.[68]

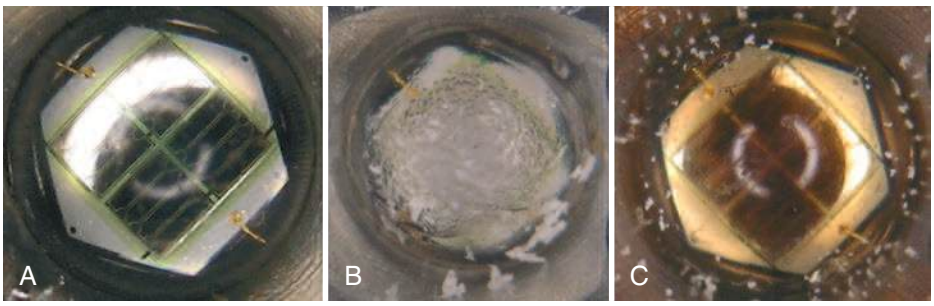
Second-Generation LED Lights

Advancements in LED chip design and increased radiant power output provided small, integrated packages containing surface area emitting LED pads instead of discrete LED casings. These new chips greatly increased the radiant power output of curing lights to the level that the number of photons emitted within the absorption range of CQ increased, and sometimes exceeded, what was delivered from high-output QTH or PAC lights.[1] Thus the recommended exposure times from some curing lights using the new high output blue LED chips became less than those for QTH lights, a feature that further increased the marketability of LED curing lights. However, with the increase in radiant power output, the need to cool the LED chip became critical; otherwise the LED would overheat and destroy itself. Fig. 6.19 displays images of such damaged arrays. For this reason, internal cooling fans as well as large metal heat sinks were used within these units. Often the metal cladding of these devices also acted as a large heat sink. Fig. 6.20 shows an example of a LED chip assembly, heat sink, and cooling fan. The typical second-generation blue LED light still has a relatively narrow emission spectrum with a full-width half-maximum (FWHM) emission spectrum that is still only about 25 nm (Fig. 6.21).

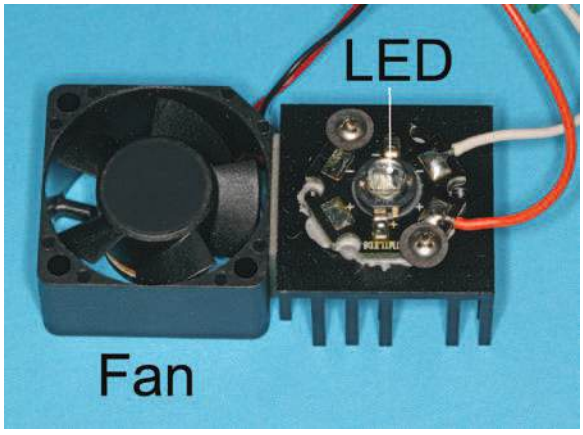
These curing lights were more compact and more powerful than the "first generation" of LED lights and thus were termed the "second generation" products. However, neither the first- nor the second-generation LED curing lights delivered much light of wavelengths below 420 nm, and thus they were incompatible with some of the "alternative" photoinitiators, such as Lucirin TPO.

Third-Generation LED Lights: Multiwave, Multipeak, Polywave

As vital, intraoral tooth bleaching became popular, many dentists discovered that the shades of composites available at that time were too yellow and did not match the color of bleached teeth. This effect occurred because the CQ photoinitiator used in most dental resins is bright yellow. To overcome this problem, some manufacturers incorporated co-initiators to boost the effectiveness of CQ and thus reduce the amount of CQ required in the resin, while others started using resin compositions that contained alternative photoinitiators to reduce the CQ concentration.[1] These alternative initiators imparted less of a yellow color in the resin and allowed manufacturers to make translucent and bleaching (lighter) shades of their restorative resins. These photoinitiators still function within the visible spectrum, but they are activated by shorter wavelengths of light closer to violet (<420 nm) light and could be photocured using the broad emission spectrum from QTH and PAC curing lights. Unfortunately, the first- and second-generation LEDs that emitted only blue light above 420 nm could only activate the CQ component within these resins and were unable to activate the alternative initiators.



• **Fig. 6.19** Undamaged LED chip (A) compared to overheated and damaged LED chips (B, C).



• **Fig. 6.20** Example of the internal layout of a second-generation LED curing light having a single emitting LED chip. Note the heat sink and fan that help dissipate heat away from the LED chip.

To solve the problem caused by the narrow emission spectrum from the blue-only LED units, an additional color chip(s) was added to the blue LED array (Fig. 6.22). By incorporating different LED color chips into the curing light, the lights now emitted both the lower wavelengths of violet light, usually from 390 to 430 nm, as well as light from 440 to 500 nm[1,2] and could now activate all the different photoinitiators used in a dental resin.

These LED curing lights have been called the "third generation" units, meaning that they emit light of more than one wavelength range (or color). Curing lights of this generation have also been described as "multiwave," "multipeak," or *Polywave* curing lights. The term *Polywave* is a registered trademark of Ivoclar Vivadent, and applies to their current line of "Bluephase" curing lights. The emission spectrum from a typical third-generation LED light is shown in Fig. 6.23, where it can be seen that the emission spectrum from these units includes wavelengths that activate not only CQ but also the alternative initiators.[1,2] However, including radiant power in the lower wavelengths may not be ideal for photo-curing bulk-filled resin-based composites because of the limited ability of the shorter wavelengths of violet light to reach down to the bottom of 4 or 5 mm of resin composite. Here the resin will polymerize primarily due to the activation of the CQ photoinitiator by the longer wavelengths of blue light. Additionally, although these LED units produce a broader emission spectrum, in some lights, the location of the various different wavelength LED emitters in the LCU affects both the irradiance beam profile and the wavelength distribution of light from these curing lights.[69,70] This finding is concerning because a strong positive correlation has been reported between the irradiance beam profile values and localized microhardness values both across and through the resin composite.[69,71,72] For this reason, the internal optics of broad-spectrum, multiple peak LED curing lights that include multiple LED emitters must be carefully designed so that a homogeneous light output is produced.

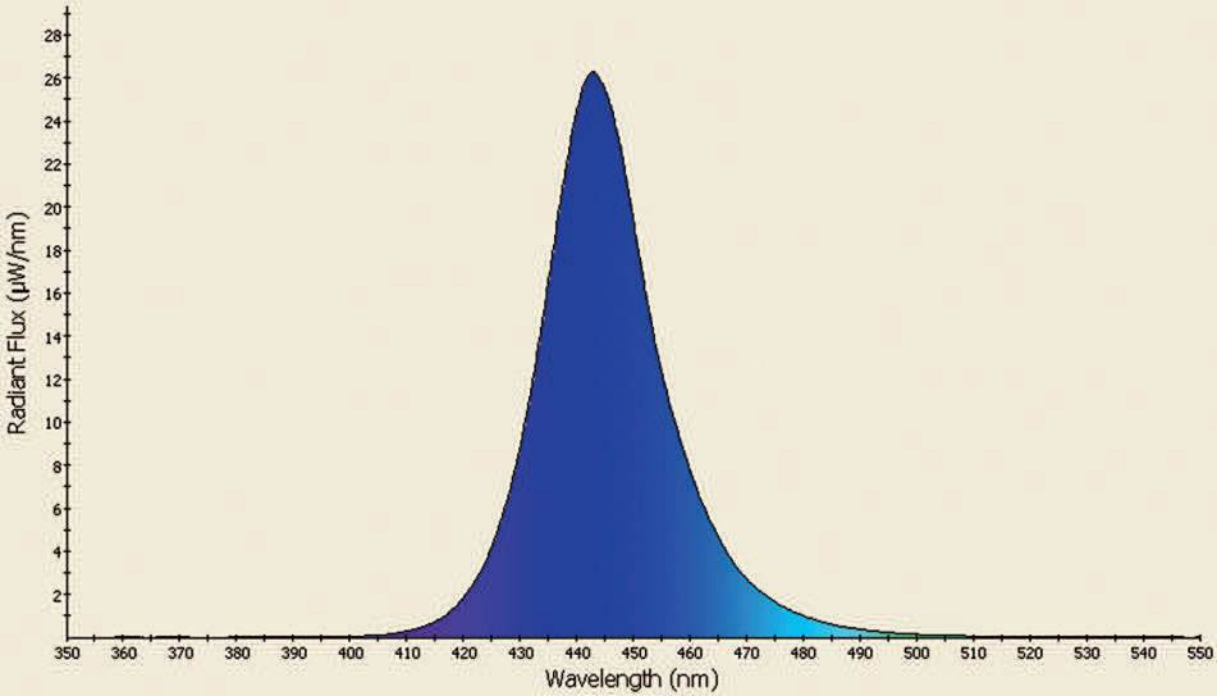
To appreciate differences in the spectral distribution of photons emitted from different types of dental curing lights, Fig. 6.24 shows examples of the emission spectra from QTH, PAC, and LED curing lights. Note the relatively narrow emission spectrum from the LED units and the high spectral radiant power emitted by LED units in the 450-nm region compared to the broader range of wavelength emitted by QTH and PAC lights.

Batteries

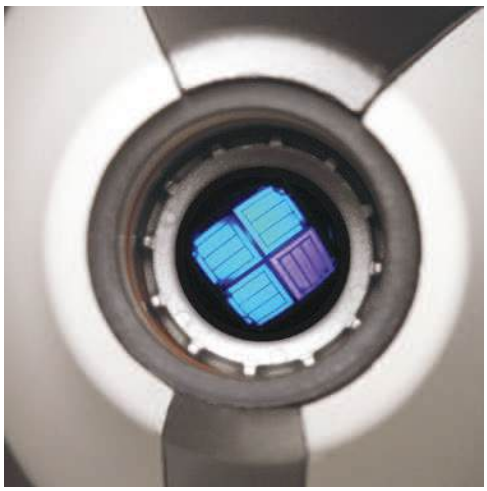
Originally, LED curing lights were powered by nickel-cadmium (NiCad) batteries. These batteries can be damaged if they are discharged to below 75% of their rated voltage, or by being overcharged. Smaller, high-capacity, nickel-metal hydride or lithium-polymer batteries later replaced the NiCad sources, and these newer types should last between 2 to 3 years from the date of manufacture before needing to be replaced. Ultracapacitors are well-suited for applications such as a curing light, where high bursts of power are required for a short time. They are used in one curing light to deliver sufficient power for approximately 25 ten-second exposures before needing to be recharged. However, ultracapacitors can be fully recharged in as little as 40 seconds and they may last up to 10 years before needing to be replaced.

Budget Curing Lights

The dental profession can now purchase dental equipment, including curing lights, directly online and at low cost, without using an approved distributor. These "budget curing lights" may not have been approved for sale in the country of use, and often use small diameter (6 to 7 mm) light guides. As described previously in the chapter, by decreasing light tip diameter, budget curing lights can deliver the same irradiance values as more expensive lights from major manufacturers that use wider diameter tips (Fig. 6.25). The light beam profiles from these budget lights can also be inferior compared with those of the higher priced lights from major manufacturers,[20,21] and the electronics in these units may not compensate for the decrease in the output from the battery during use. Thus, without any warning to the user, the light output from battery-operated budget curing lights may not be stable during operation.[20] Dental curing lights are medical devices; thus it is recommended that you do not use curing lights that have not been approved for use in your country.



• **Fig. 6.21** Emission spectrum from a typical second-generation LED light (Bluephase 16i, Ivoclar Vivadent). Note the single emission peak in the blue wavelength range.



• **Fig. 6.22** Multiple wavelength emitting LED chip set in a commercial third-generation *Polywave* light (Bluephase G2, Ivoclar Vivadent) showing an array of three blue chips and one violet LED chip in the body of the light. Note the absence of any filter within the unit.

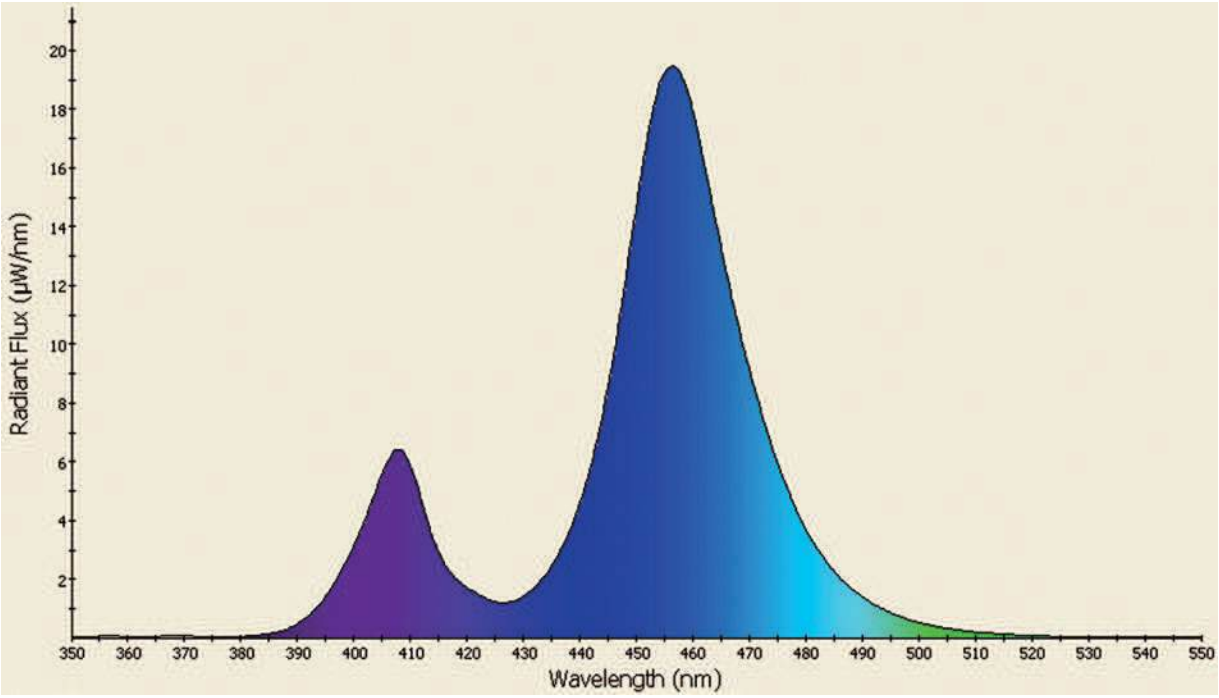
Evaluation of Light Output From a Curing Light

Radiant Power Measurement

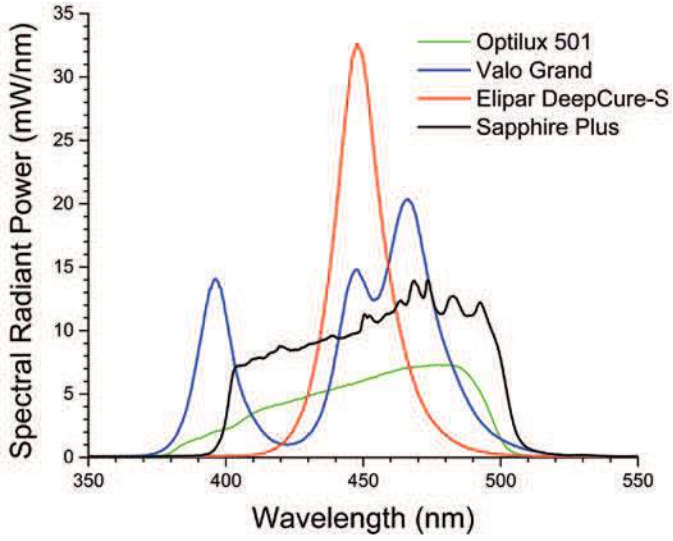
The 2015 ISO 10650 standard for measuring the output from dental curing lights recommends using a laboratory-grade power meter to measure radiant power output and dividing this power by the tip area to produce an average irradiance value from the curing light.[73] An example of a laboratory-grade power sensor is a thermopile that converts thermal energy delivered by photons into electrical energy. The sensor is composed of hundreds of interconnected thermocouples that generate an output voltage that is proportional to a local temperature difference. An alternative power sensor also suggested in the ISO 10650 standard is the PowerMax-Pro 150HD with a broadband detector (Coherent Inc., Santa Clara, CA). This detector differs from a thermopile (Fig. 6.26) because the sensor contains a stack of thin films to generate an electrical current when light hits the surface of the detector. In contrast to thermopiles that can take several seconds to respond, this type of sensor responds rapidly and can provide a radiant power reading within microseconds.[21,74]

Spectral Radiant Power Measurement

An optical spectrometer, or more accurately termed a spectrophotometer, measures the properties of a light source over a specific portion of the electromagnetic spectrum. Inside the spectrophotometer, light is broken down into its various wavelength components, and the signal is digitized as a function of wavelength and power (W/nm). The first step in this process is to direct light from a source being evaluated into a spectrophotometer through a narrow aperture known as an entrance slit. This slit vignettes the light and directs it onto a grating. The grating inside the spectrophotometer then disperses the various spectral components of the light, much like a prism, at slightly varying angles, onto a detector. These gratings can be ruled or holographic diffraction plates. Holographic gratings produce less stray light, while ruled gratings tend to be more reflective. The groove density on a grating determines the extent of light dispersion, and the optimum spectral range (bandwidth) is a function of the groove density. The greater the groove density, the better is the optical resolution, but the more truncated will be the spectral range. Thus, to produce the best response from the spectrophotometer, the correct slit dimensions



• **Fig. 6.23** Emission spectrum of a typical third-generation *Polywave* LED curing lights (Bluephase Style, Ivoclar Vivadent). Note the distinct emission peaks in the violet and blue wavelength ranges.



• **Fig. 6.24** Emission spectra from a PAC (*black line,* Sapphire DenMat), QTH (*green line,* Optilux 501, Kerr), and second-generation (*red line,* Deep Cure-S, 3M), and third-generation (*blue line,* Valo Grand, Ultradent) LED curing light.

and grating groove density must be carefully chosen to match the radiant power, the wavelength being examined, and the resolution required. A schematic of the internal components of a miniature spectrophotometer is shown in Fig. 6.27.

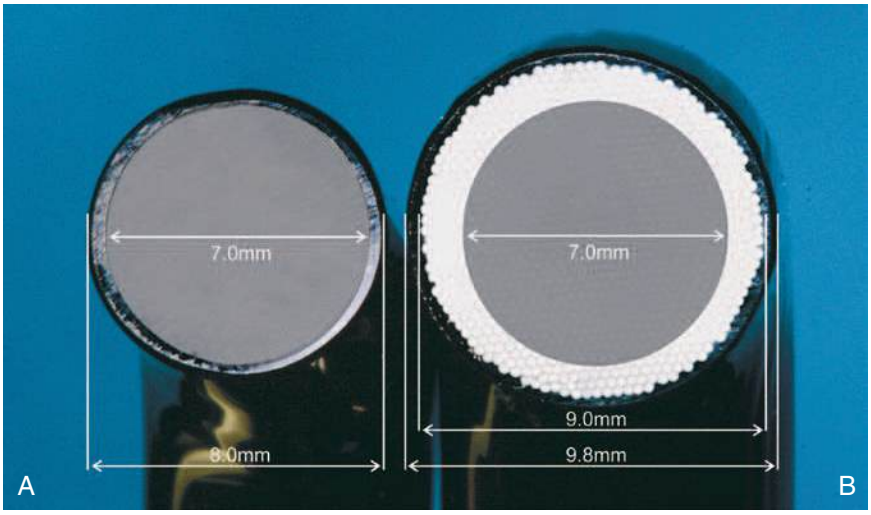
Integrating Sphere

The total radiant power (radiant flux) from a light source can be collected, without any influence of the directional characteristics from the source, using an integrating sphere (also known as an Ulbricht sphere). This device is a hollow sphere that has a diffuse, white, reflective inner coating. Light enters the sphere through a port and strikes the sphere sides. The light is then scattered in a diffuse manner, called Lambertian reflectance, that minimizes the effects of the original direction of light. Thus only light that has been diffused within the sphere reaches the detector. This is usually via a fiberoptic cable that is connected to the entrance port of a spectrophotometer; however, the detector can also be attached directly to the sphere. Integrating spheres can be small and portable (Fig. 6.28) or they can be so large that a person can walk inside.

Light Beam Uniformity

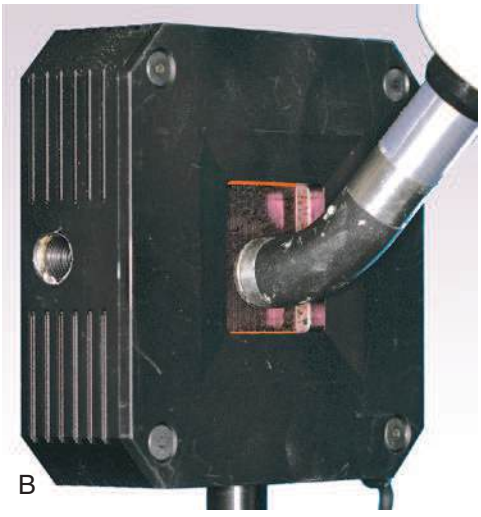
Recognizing that the power and emission spectrum across the light tip cannot be adequately described by a single value, laser beam analyzers have been used to examine the homogeneity of power and the wavelengths of light emitted from curing lights. The charge couple detector (CCD) in a digital camera is used to record an image of the light as the light strikes a diffusing target screen. Since the type of screen or the use of a bandpass filter can affect the image, these components must be chosen carefully.[70] The radiant power received by each camera pixel in the CCD array is recorded and displayed as a color-coded, two- or three-dimensional image of the light output across the light tip end. A typical arrangement for using a beam analyzer to characterize the light distribution across the emitting end of a light-curing tip is shown in Fig. 6.29.

Several publications have shown that the irradiance from many dental curing lights is not evenly distributed across the emitting end of the light tip.[20,21,69,70,71,72,75,76] In addition, because of how the light is produced, the emission spectrum is not evenly distributed across the emitting end of the light tip of some broad-spectrum LED curing lights.[70] In contrast, although the local irradiance values may differ greatly, because of the light source used, the emission spectra are distributed evenly across the tip for QTH,

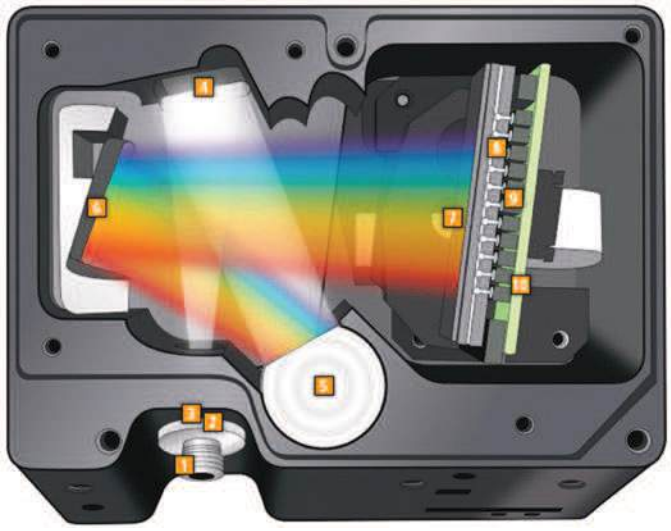


• **Fig. 6.25** By decreasing the internal tip diameter from 9 mm to 7 mm, the output area decreases. If the same radiant power is emitted, the irradiance will increase when using the 7 mm diameter light tip. 9 mm diameter = area of 63.6 mm^2, 400mW radiant power: irradiance = 629 mW/cm^2; 7 mm diameter = area of 38.5 mm^2, 400mW radiant power: irradiance is now increased to 1039 mW/cm^2.





• **Fig. 6.26** Images of a conventional thermopile (A) and a PowerMax Pro 150HD with a high-speed broadband sensor (B).



• **Fig. 6.27** Schematic diagram of a spectrometer. 1. Fiberoptic connector. 2. Interchangeable entrance slit. 3. Long pass filter. 4. Collimating mirror. 5. Grating. 6. Focusing mirror. 7. Collection lens. 8. Detector. 9. Variable long pass filter. 10. UV detector window. (Image courtesy Ocean Optics.)



• **Fig. 6.28** A small integrating sphere attached to a miniature fiberoptic spectrometer (Ocean Optics).

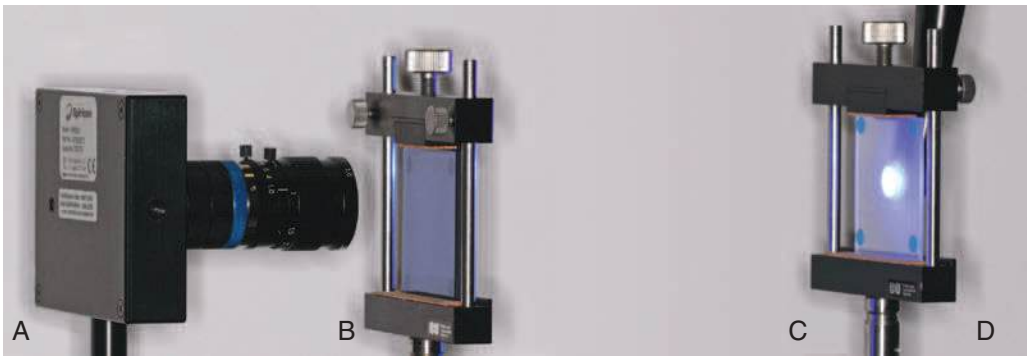
PAC, and single peak LED units.[76] The localized inhomogeneity in the irradiance results in "hot spots" of high-intensity light and "cold spots" of lower values, where very little light, or different wavelengths of light, may be present. Examples of homogeneous and inhomogeneous distributions of light across the emitting tips of different dental curing lights are shown in Fig. 6.30.

The beam profile images illustrate that the irradiance across the light tip is a range of values and the irradiance homogeneity depends on the design of the curing light. Thus using a single irradiance value to describe the output from a curing light does not adequately describe the irradiance that is delivered across the entire light tip.[76] Instead, this commonly used value represents the average irradiance value that is emitted across the tip.

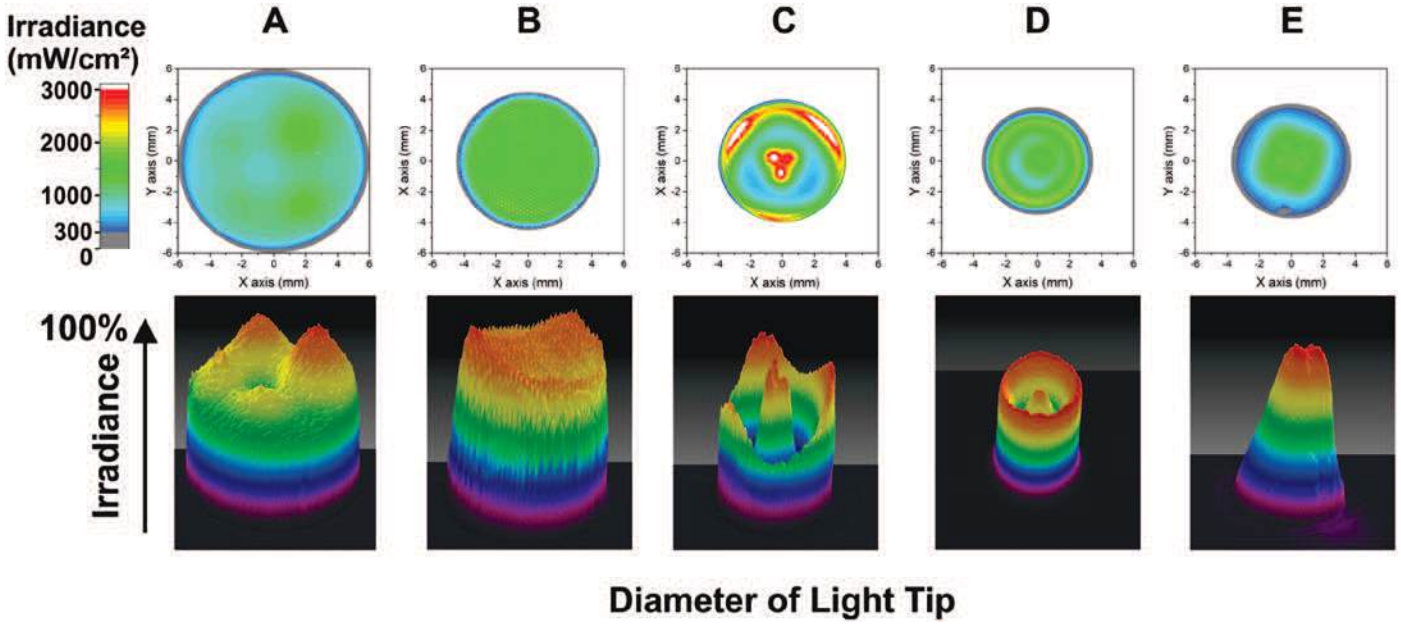
Fig. 6.31 shows how scaled images of the irradiance beam profile can be overlaid on an image of a tooth preparation to demonstrate the local irradiance values that are delivered to different locations of the preparation. These beam profile images show that, unless the operator is careful, the irradiance and radiant exposure received at the gingival margins of the proximal boxes from some curing lights may be inadequate to ensure adequate polymerization of the adhesive or the resin restoration in these regions.[20,21,69] These regions are also the most difficult to reach and are the furthest away from the light source.[45,77] Consequently, the resin at these locations may well be underpolymerized unless the operator pays careful attention and ensures that all areas of the resin receive an adequate amount of light.

Monitoring the Output From Curing Lights

When using a curing light, the intense blue light and the hardness of the topmost layer of resin provide a false sense of security that the curing light has adequately polymerized all of the restorative material that is underneath the top, exposed surface. Without warning the user, the light source inside the unit can degrade over



• **Fig. 6.29** Typical arrangement for using a beam analyzer to examine a dental curing light: (A) the CCD camera, and neutral density filter (B) and the diffusing screen (C) upon which the output of the curing light tip is directed (D).

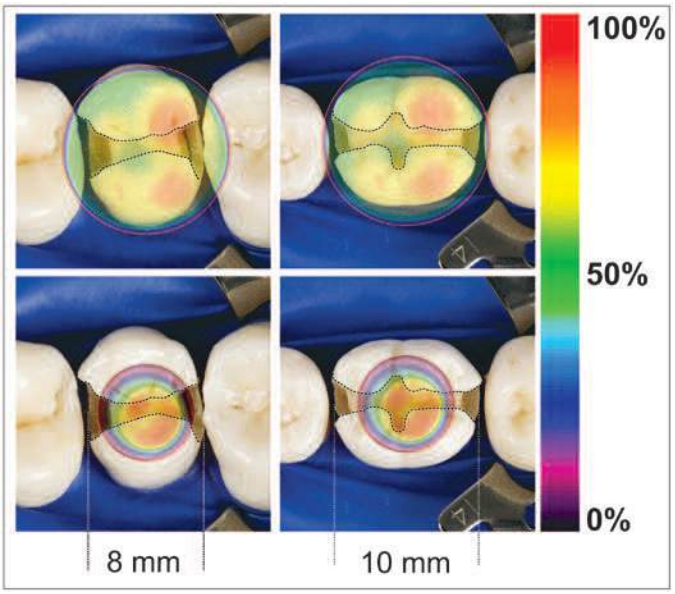


• **Fig. 6.30** Images of beam profiles from five contemporary curing lights. The upper set of images are a scaled, two-dimensional representation of the irradiance distribution from 0 to 3,000 mW/cm^2. The lower view is a three-dimensional representation with all images scaled to their individual maximum irradiance value. Note the differences in the beam diameters and the inhomogeneous irradiance distribution across the light tips. Using a single irradiance value cannot fully describe the light output.

time[20,78] and thus decrease the light output. The light output can also be reduced due to the buildup of scale on the fiberoptic light probe after repeated autoclaving,[79] after breakage or fracture at the light tip,[80,81] or due to the presence of cured composite resin or debris on the light tip.[80,82] All these factors can significantly decrease the ability of the curing light to photocure the resin (Fig. 6.32). Thus it is important to use a dental radiometer to routinely measure the output from the curing light. The value should be recorded, as well as the condition of the curing light. Ideally, the value of the light/tip combination should be recorded when the unit is new and subsequent readings can be used to indicate the relative performance of the curing light as a function of time.

Handheld "Dental Radiometers"

Examples of seven handheld dental radiometers that can be used to monitor the output from dental curing lights are displayed in Fig. 6.33A. These radiometers usually have a bandpass filter in front of a silicon photodiode detector that converts the photons received from the curing light into electrical current and, when calibrated, into units of irradiance. However, the light is rarely emitted uniformly across the entire light tip, and because the entrance apertures on some dental radiometers are smaller than the light tip (see Fig. 6.33B), most dental radiometers do not

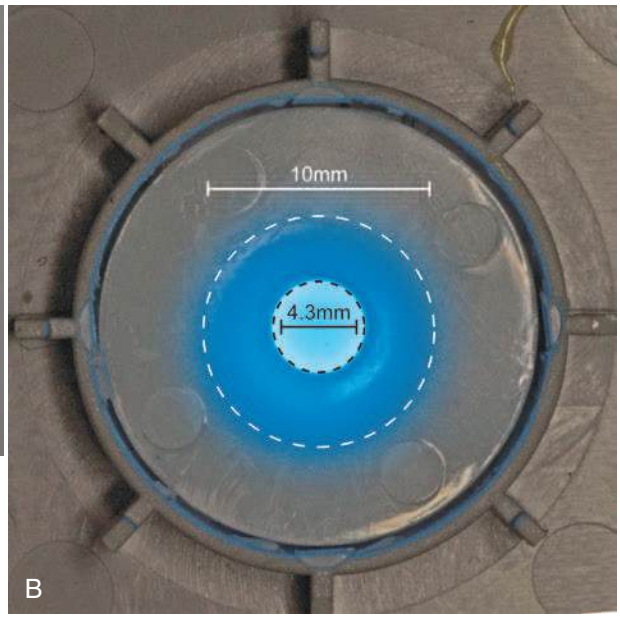


• **Fig. 6.31** Images of good *(top)* and poor *(bottom)* beam profiles, superimposed on images of premolar and molar tooth preparations. Each image is normalized and scaled to 100% of its individual, maximum irradiance value. (From Rueggeberg FA, Giannini M, Arrais CAG, Price RBT: Light curing in dentistry and clinical implications: a literature review, *Braz Oral Res* 31(suppl 1): e61, 2017. http://dx.doi.org/10.1590/1807-3107bor-2017.vol31.0061. Licensed under CC BY 2.0, https://creativecommons.org/licenses/by/2.0/.)

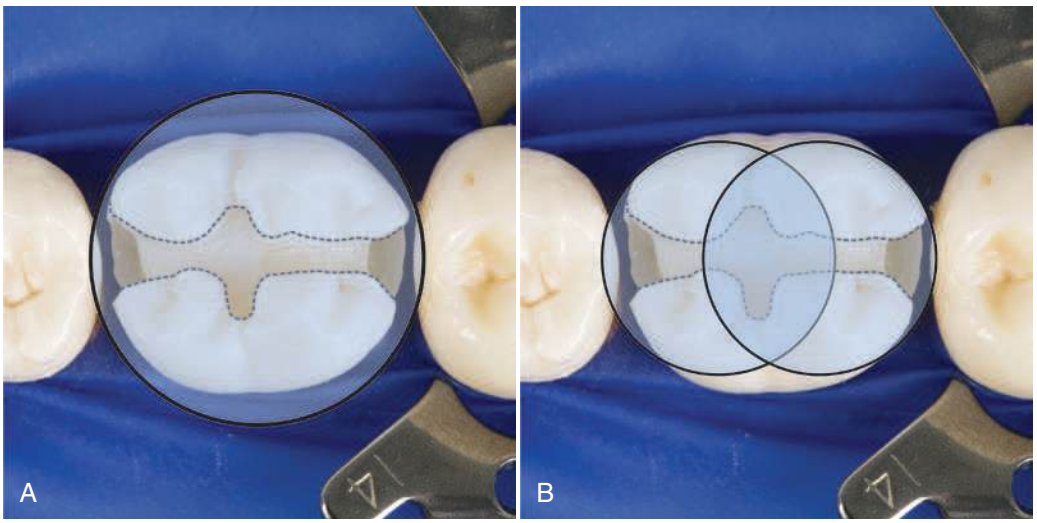


• **Fig. 6.32** Examples of damaged (A) and debris-contaminated light tips (B and C).





• **Fig. 6.33** Image of seven commercial dental radiometers (A), and an example of the difference between the 4.3-mm entrance aperture into the radiometer and the 10-mm tip of the curing light (B).



• **Fig. 6.34** A large diameter light tip (A) can cover an entire molar tooth, whereas smaller diameter light tips (B) require multiple exposures to cover an entire the surface of a molar MOD restoration.

capture all the light from the light emitting tip.[83] Thus, depending on where the light tip is positioned over the detector area, a region of high or low irradiance may be measured. This limitation helps to explain why several studies have reported that most dental radiometers are inaccurate.[83-86] In addition, because QTH, PAC, or LED dental curing lights all deliver different emission spectra, the type of bandpass filter used within many dental radiometers will affect the accuracy of the irradiance value reported.[83,85] Thus the radiometer manufactured by the maker of the curing light being tested should be the most accurate.

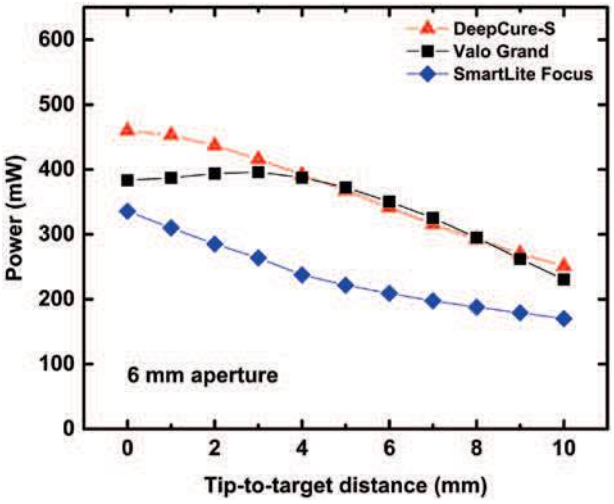
Practical Considerations for Light Curing Dental Resins in the Mouth

Factors Affecting Light Delivery to the Target

Table 6.1 shows that when the irradiance delivered is multiplied by the exposure time, the product describes the radiant exposure (in J/cm^2) delivered to the exposed surface of the resin. It has been recommended that a QTH curing light should deliver a minimum irradiance of 400 mW/cm^2 for 60 seconds to adequately polymerize a 1.5- to 2-mm-thick increment of resin.[87] It has also been recommended that most dental resins require a radiant exposure of approximately 16 J/cm^2 to adequately cure to a depth of 2 mm.[88] Thus this threshold could be reached in 40 seconds if the curing light delivers an average irradiance of 400 mW/cm^2 to the resin. However, different brands of resin require different exposure times, and the minimum radiant exposure may range from 6 to at least 24 J/cm^2 to achieve an adequate level of polymerization in the various shades and colors.[89,90]

Light Guide Tip Design

As discussed previously, the design of the light tip can have a significant impact on the irradiance delivered to the restoration[45,91] and can negatively affect the physical properties of the resin and its bond strength to the tooth.[7,92-94] Twenty years ago, most light tips had a 10 to 13 mm tip diameter, but recently this has changed and many light tips are now only 6 to 8 mm in diameter. Although a smaller diameter tip can increase irradiance, small light tips may not cover the entire restoration area. Since there is increased interest in bulk filling and bulk curing of large restorations, this decrease in tip diameter is now of concern. With the dimensions of a mandibular molar being approximately 11.0 mm mesiodistally and 10.5 mm buccolingually at the crown,[95] the clinician who wishes to reduce time spent light curing restorations should use a curing light with a tip that completely covers the entire restoration surface. Otherwise, as seen in Fig. 6.34, even when using a resin intended for bulk filling, multiple, sequential, overlapping exposures will be required to ensure that all areas of the restorative material have received an adequate amount of light.[21]



• **Fig. 6.35** Overlay graph indicating differences among three commercial curing lights with respect to the power received using tip-to-target distances from 0 to 10 mm.

Distance to Target

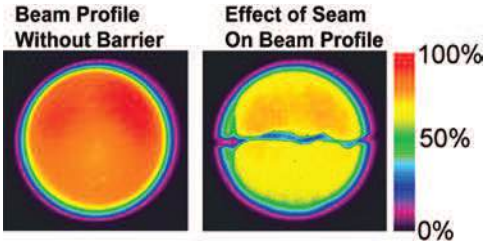
Irradiance (radiant exitance) values stated by manufacturers are usually measured at the light tip end. Because the resin is usually 2 to 8 mm away from the light tip, this value does not reflect the irradiance values that are received by the restoration. Fig. 6.35 displays how the effect of increasing the tip-to-target distance differs



• **Fig. 6.36** Examples of different types of plastic barriers placed over the curing light to minimize contamination. Ideally, the plastic barrier should cover the entire unit instead of just the light tip.



• **Fig. 6.37** The barrier should fit snugly over the light tip and the seam should be positioned so that it does not impede light output.



• **Fig. 6.38** If the seam of the barrier crosses the light tip, light output will be reduced.



• **Fig. 6.39** Example of a curing light with air vents that should not be covered by the protective barrier.

among various brands of dental curing lights. Depending on their design, some curing lights deliver only 25% of the irradiance measured at the tip when they are 8 mm away from the resin surface.[7,45,96-98] This decrease occurs because the design and the optics of the light source and the light guide all affect the beam divergence angle from the distal tip end of the light guide. Some of the light from a curing light is delivered as a collimated beam, some is focused to a point, and some is immediately dispersed laterally from the tip. Thus curing lights do not act as a point source and depending on the unit's optical design, which sometimes includes a lens, the effect of changes in the tip to target distance on the irradiance received does not obey the inverse square law.

Infection Control

Although some fiberoptic light guides can be autoclaved, the body of the curing light itself cannot. Thus the use of infection control barriers, such as the plastic barriers presented in Figs. 6.36 and 6.37, that ideally should cover the entire curing light and light guides are recommended. The barrier should fit snugly over the light tip and the sleeve seam should not impede the light output. Fig. 6.38 illustrates the detrimental effect that occurs when the seam of a barrier traverses across the light tip and disrupts the light beam.

Unfortunately, the design of these preformed plastic sheaths that fit over a curing light is not standardized and they may not be designed to optimize light transmission (Fig. 6.39). Some barriers can reduce the radiant exitance by up to 40%, and latex-based barriers should be avoided as they have been reported to negatively affect resin polymerization.[75,99-101] For this reason, the output from curing lights should be tested on the radiometer with the barrier over the light tip. Clear, plastic food wrap stretched over the light tip in a single layer can also be an effective and inexpensive infection control barrier that has minimal effect on light output.[81,100,101]

When using cold sterilizing techniques, only approved cleaning solutions should be used. Disinfectant sprays can erode the O-rings used to stabilize light guides, and the residual fluid can bake onto the lens or reflectors inside the light housing, thus decreasing the light output.[102] Therefore, if it can be detached, the light guide should be removed from time to time and any lens, reflector, and filter inside the curing light should be checked to ensure that they, and both ends of the light guide, are clean and undamaged.

Health-Related Issues

Intrapulpal Temperature Considerations

An *in vivo* study involving the direct application of heat from a soldering iron to the facial surfaces of intact rhesus monkey teeth indicated that an increase in intrapulpal temperature of 5.5°C resulted in pulpal necrosis in 15% of the teeth tested.[103] This temperature value has been extrapolated to predict that similar pulpal temperature increases when light curing may cause pulpal necrosis to an already affected human dental pulp.

As the radiant power output from curing lights has increased, the potential for generating a damaging temperature increase in the pulp and oral tissues has also risen.[6,68,104-108] This increase in temperature is related to the amount of energy delivered (the product of irradiance delivered and exposure time) and the thermal properties (conductivity and diffusivity) of the tooth. Based on *in vitro* test results, it is suggested that curing lights that deliver an irradiance greater than 1,200 mW/cm^2 should be used for at most 15 seconds.[108]

However, this guideline has been questioned because recent human-based intrapulpal temperature studies have measured the baseline temperature of anesthetized, healthy human pulp tissue.[109] In that work, patients requiring extraction of bicuspids for orthodontic reasons volunteered as participants. Class I preparations were made, which intentionally exposed the most coronal portion of the buccal pulp horn. A sterile temperature probe was inserted directly into the pulp chamber and remained in place during temperature measurements, while the access opening was filled with temporary restorative material. The intrapulpal temperature rise in the teeth, when using a LED curing light, was then measured. The radiant exposure was determined, and the resulting increase in intrapulpal temperature is shown in Fig. 6.40.

As the radiant exposure increased, there was a corresponding increase in intrapulpal temperature. However, only when the radiant exposure levels approached 72 J/cm^2 did some *in vivo* temperature values meet or exceed the 5.5°C increase threshold, and thus potentially cause pulpal damage.[103]

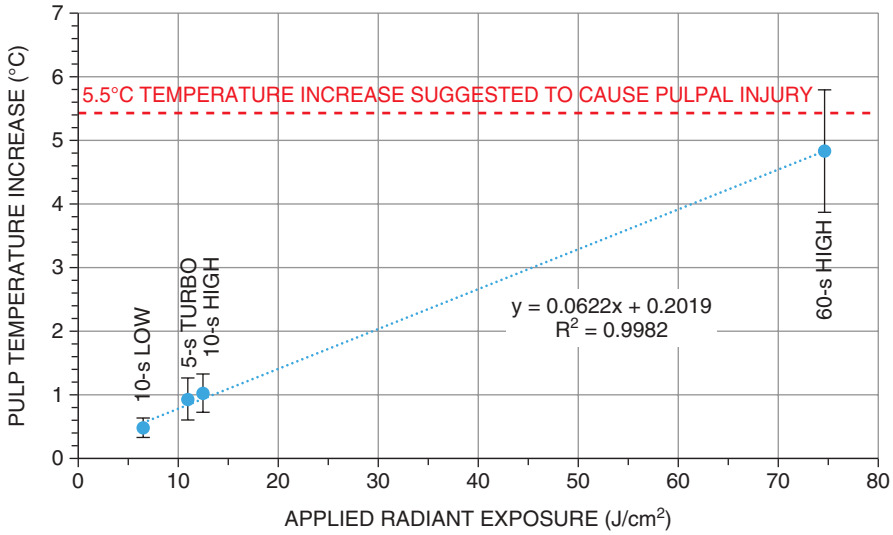
Time-based profiles of *in vivo* temperature values inside the pulp of upper bicuspid teeth exposed using the same curing light, but with one tooth having an intact buccal enamel surface and another tooth having a Class V preparation with approximately 1-mm thickness of dentin remaining, are shown in Fig. 6.41.

Although both conditions demonstrate similar pulpal temperature prior to the 30-second exposure, the rate and extent of temperature increase is much greater in the tooth with the Class V preparation than in the intact tooth. In addition, it takes longer for the pulpal temperature to return to baseline levels at the end of the exposure. Such evidence supports the concept that the potential for causing temperature increase in pulpal tissue greatly increases as more tooth structure is removed and there is less dentin overlying the pulpal tissues.[110,111]

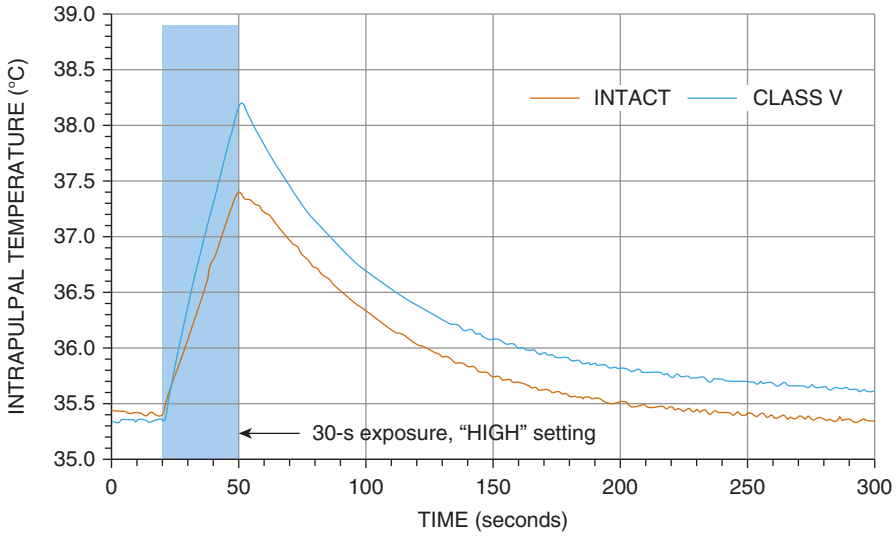
Where the pulp is at greater risk, such as in deep cavities with little overlying dentin, consideration should be given to the choice of LCU and light exposure program. It has been recommended that the tooth should be air-cooled during light exposure[112,113] from the opposite side (Fig. 6.42) of the exposure. Alternatively, a high-volume suction device can be used to cool the tooth by pulling air across the coronal portion of the tooth.

Fig. 6.43 illustrates the *in vivo* cooling effect of the air spray when it was applied 3 seconds prior to light exposure, and stopped when the light was turned off.

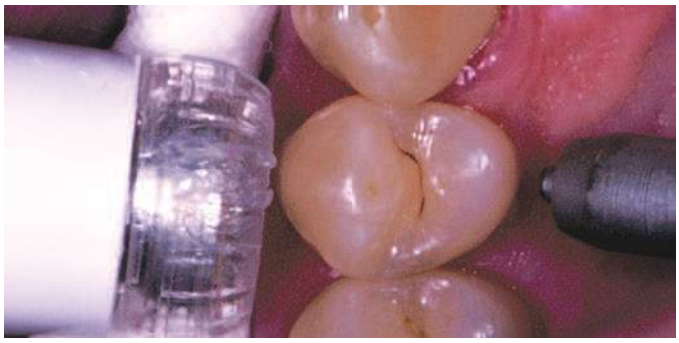
Although the pulpal temperature starts to rise shortly after light exposure, when air-cooled, the pulp temperature decreases both during and after light exposure. Consequently, it may be possible to use irradiance levels greater than 1200 mW/cm^2 for longer than 15 seconds if appropriate cooling measures are used.



• **Fig. 6.40** Relationship between delivering known radiant exposures to the facial surfaces of intact, upper bicuspid teeth and the resulting increase in human intrapulpal temperature rise when using a commercial *Polywave*® LED-based curing light. Exposure duration and output mode selection noted vertically for each group. N = 15 per test group. (Courtesy Dr. Cesar A.G. Arrais.)



• **Fig. 6.41** Comparison of real-time temperature profiles inside the pulp of an intact bicuspid, and one having a Class V preparation with approximately 1-mm of dentin remaining when exposed to the same output time and curing mode from an LED-based curing light. (Courtesy Dr. Cesar A.G. Arrais.)



• **Fig. 6.42** Directing an air syringe at the opposite surface being light-exposed will help to lower the *in vivo* intrapulpal temperature. (Courtesy Professor Marcelo Giannini.)

Soft Tissue Damage

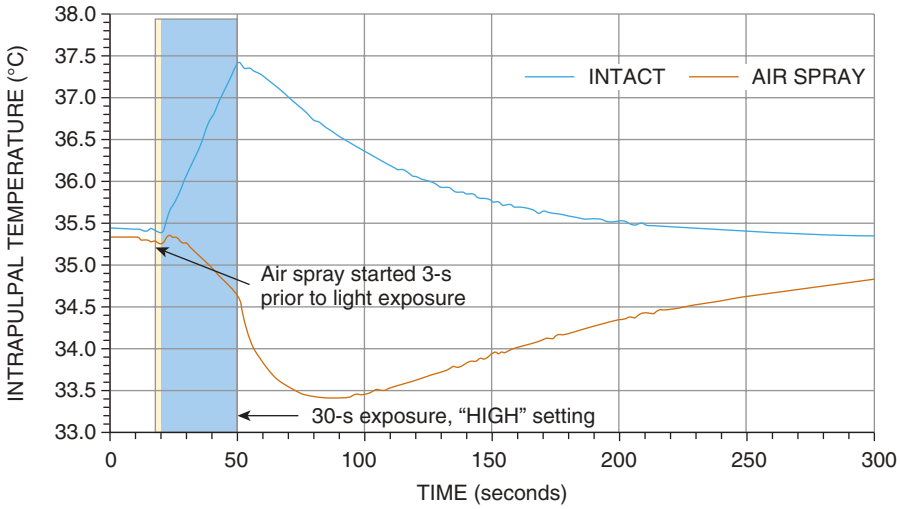
Current literature has reported several cases of soft tissue burns that were caused by excessive heat either from a hot light curing tip or from excessive amounts of light energy delivered to the oral tissues.[114] Soft tissue damage to oral tissues arising from photocuring restorations can arise from two sources: heat developed by the external housing of the curing light or from heat developed in tissues that are exposed to the high-intensity light. Often the light guide of the curing light is used to retract the cheek or tongue. For LED-based lights that contain the chip sets at the tip, heat generated by the chips is diffused to the outer surface of the curing light. In so doing, the surface temperature of areas on the curing light body opposite to where the chips are emitting may become hot and thus burn the soft tissues. Fig. 6.44 provides examples of how the distal portions of light guides are used to retract the soft tissue and illustrates the potential for causing soft tissue damage. However, the maximum acceptable external curing light temperature increases for curing lights is stipulated in the International Electrotechnical Commission specification, IEC 60601, that applies to dental curing lights.[115] Thus, if the curing light has passed this specification, the heat from the external housing of the unit should not be high enough to cause a burn.

When light-curing gingival areas, as in Class V situations, the gingival portions of veneers, and the margins of crowns, a portion of light output also reaches soft tissue that is often not protected by a rubber dam. Because of the translucent properties of the poorly keratinized oral epithelium, light easily penetrates and is absorbed by these tissues. This may result in a significant tissue burn if some protection is not provided. Even when using a rubber dam, the potential for elevated tissue temperatures may occur.[114] Fig. 6.45 displays a tissue burn resulting from a 30-second direct light exposure from a high-intensity LED curing light on the attached gingiva of a pig. Thus, especially when using contemporary high-power curing lights, clinicians must keep in mind the potential for causing both direct heat and radiant thermal damage to the soft tissues and the dental pulp. Clinicians should closely monitor the position of the light tip and not arbitrarily increase the exposure time in an attempt to ensure maximal polymerization without using strategies to minimize thermal damage from the curing light. One option is to place a small section of cotton gauze under the dam to protect the gingiva (Fig. 6.46). Another is to carefully position the light tip just over the resin and watch what you are doing when light curing so that the light tip does not wander over the lips or gingiva.

The Optical "Blue Light Hazard"

Dentists have a duty to protect both their patients and employees from harm. There has been concern that blue light from high-power curing lights can place dental personnel at risk for ocular damage.[116-120] Although this hazard can be prevented by using appropriate eye protection, these items are not universally used.[121]

This blue light hazard is related to the wavelength of light (Fig. 6.47). The hazard mostly occurs between about 415 nm and 480 nm and is greatest at 440 nm.[122] This wavelength range is well within the emission spectra from all dental curing lights[1] (Fig. 6.48). If eye protection is not used to eliminate the blue light, and instead the operator looks directly at the curing light, even for just 1 second during the curing cycle before averting his or her eyes, it may only take as few as seven curing cycles to exceed the recommended maximum daily exposure to blue light.[116]



• **Fig. 6.43** Comparison of real-time, human pulpal temperature values between an intact facial surface, with no air-cooling during exposure *(blue profile),* and when the lingual surface was air-cooled prior to and all throughout the light exposure *(orange profile).* Air-cooling resulted in a 2°C pulp temperature decrease. (Courtesy Dr. Cesar A.G. Arrais.)

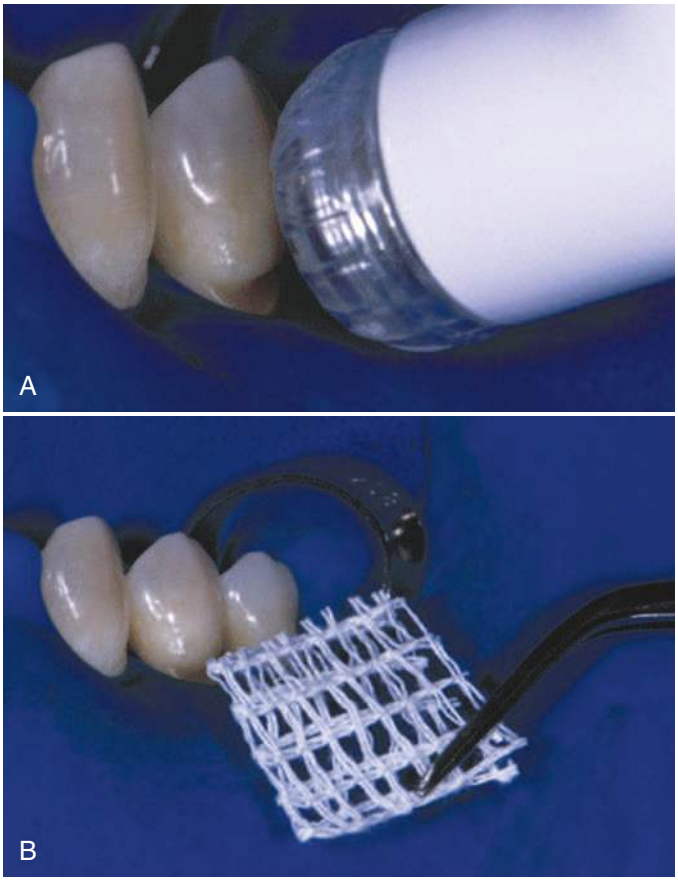


• **Fig. 6.44** Curing light used to reflect the tongue (A) and cheek (B) to gain better access to the operative site, or resting on the lip during light exposure (C).

Blue light is transmitted through the ocular media and absorbed by the retina. Although high levels of blue light can cause immediate and irreversible retinal burning, there are also concerns that repeated, chronic exposure to low levels of blue light will cause accelerated retinal degeneration and chronic photochemical injury to the retinal-pigmented epithelium and choroid. This chronic exposure is thought to accelerate age-related macular degeneration (ARMD).[123,124] Most countries follow international guidelines, such as those from the International Commission on Non-Ionizing Radiation Protection (ICNIRP) and the American Conference of Governmental Industrial Hygienists (ACGIH), when determining safe levels of exposure to blue light in all workplaces, not just

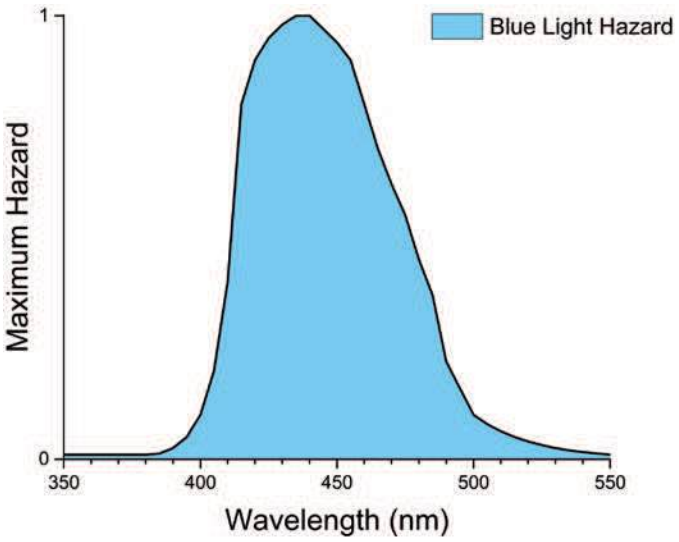


• **Fig. 6.45** Erythema resulting from exposure of pig gingiva to a high-intensity LED curing light. (Reprinted from Maucoski C, Zarpellon DC, Dos Santos FA, et al: Analysis of temperature increase in swine gingiva after exposure to a Polywave® LED light curing unit, *Dent Mater* 33(11): 1266-1273, 2017. With permission from The Academy of Dental Materials.)

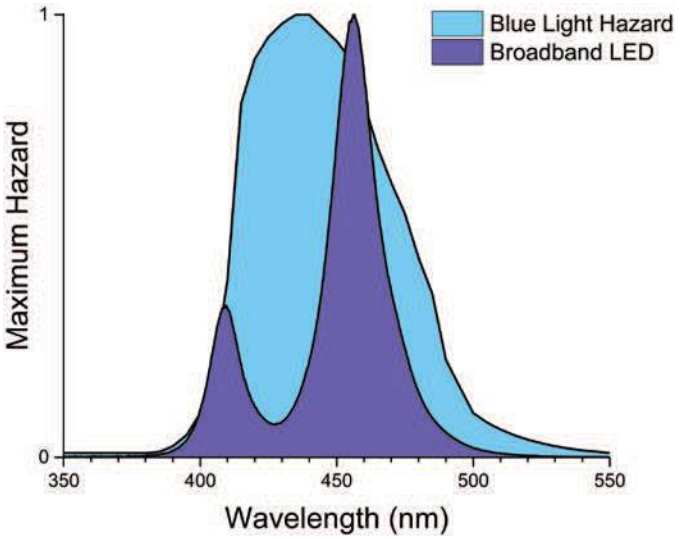


• **Fig. 6.46** Conventional positioning of a curing tip when exposing a Class V preparation (A). Such a position can cause a tissue burn even under the rubber dam. A small section of cotton gauze can be placed under the dam to protect the gingiva (B). (Courtesy Professor Marcelo Giannini.)

dental offices.[122,123] It should also be noted that these maximum recommended exposure times have been developed for individuals with normal photosensitivity. Patients or dental personnel who have had cataract surgery or who are taking photosensitizing medications will have a greater susceptibility for retinal damage.[122,123]



• **Fig. 6.47** Graphical depiction of the blue light hazard function related to wavelength. The maximum hazard values occur between 415 and 480 nm. (From Rueggeberg FA, Giannini M, Arrais CAG, Price RBT: Light curing in dentistry and clinical implications: a literature review, *Braz Oral Res* 31(suppl 1): e61, 2017. http://dx.doi.org/10.1590/1807-3107bor-2017.vol31.0061. Licensed under CC BY 2.0, https://creativecommons.org/licenses/by/2.0/.)

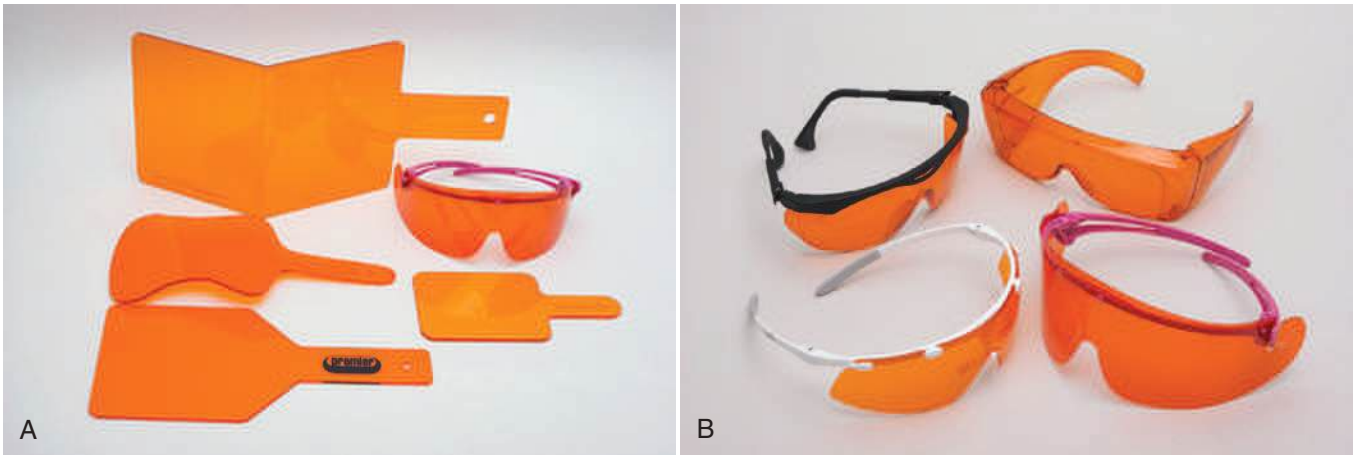


• **Fig. 6.48** Superimposition of the wavelengths of light that cause the blue light hazard *(light blue area)* and that of the emission spectrum from a multiwave LED dental curing light *(violet area).* The blue light component from dental curing light is within the hazard range. (From Rueggeberg FA, Giannini M, Arrais CAG, Price RBT: Light curing in dentistry and clinical implications: a literature review, *Braz Oral Res* 31(suppl 1): e61, 2017. http://dx.doi.org/10.1590/1807-3107bor-2017.vol31.0061. Licensed under CC BY 2.0, https://creativecommons.org/licenses/by/2.0/.)

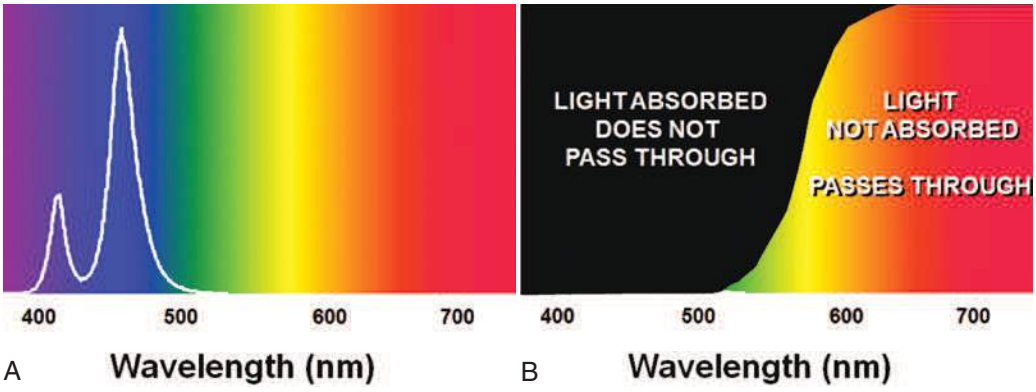
Blue Light Blocking Protection

Using appropriate ocular protection can completely prevent the blue light hazard and all LCU manufacturers supply and recommend the use of protective blue-blocking orange shields to protect the eyes from the bright blue light. These orange plastic shields are usually attached to the light tip and can be adjusted to provide eye protection to the operator. Other types of protective glasses and paddles are also available in various sizes, thickness, and designs (Fig. 6.49).

Most of the protective orange filters used in dentistry are designed to reduce the risks of exposure to wavelengths of blue light between



• **Fig. 6.49** Various forms of "blue-blocker" eye protection.



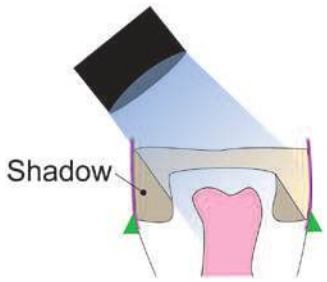
• **Fig. 6.50** Depiction of spectral emission of a multiwave LED curing light (A) and how the blue-blocker shield filters harmful radiation while allowing light of longer wavelengths to pass (B).

420 and 500 nm. A schematic representation of how blue-blocking filters block the damaging radiation from curing lights is seen in Fig. 6.50.

The best blue light filtering glasses ("orange blue-blockers") have been shown to reduce the transmission of light below 500 nm by 99%.[125] When such blue light filtering glasses are used, instead of looking away from the bright blue light, the operator can safely watch what he or she is doing when light curing; thus they can keep the light tip directly over the resin they are trying to light cure. However, not all brands of protective filters are equally effective and some protective filters have been reported to be inadequate if they are used to protect the eye against a range of wavelengths other than the intended range.[125] Thus it is important to use the correct blue-blocking filter for the curing light being used.

Electromagnetic Risk From Curing Lights

There is some concern that the electrical circuitry in curing light can interact with implanted pacemakers.[126] Although there is a theoretical risk, the initial laboratory tests were performed under conditions where the investigators used physiologic saline baths and benchtop settings. Major companies are required to test for this hazard before selling their curing lights, and there should be no risk if the curing light has been purchased from a reputable manufacturer.[127,128] There may be a risk when using untested budget curing lights.



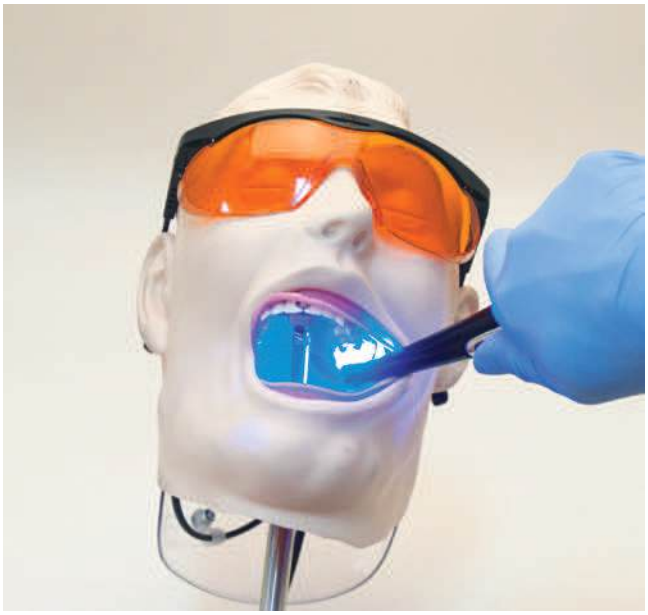
• **Fig. 6.51** The position of the light tip can produce unwanted shadows and under-cured resin.

Teaching How to Light Cure

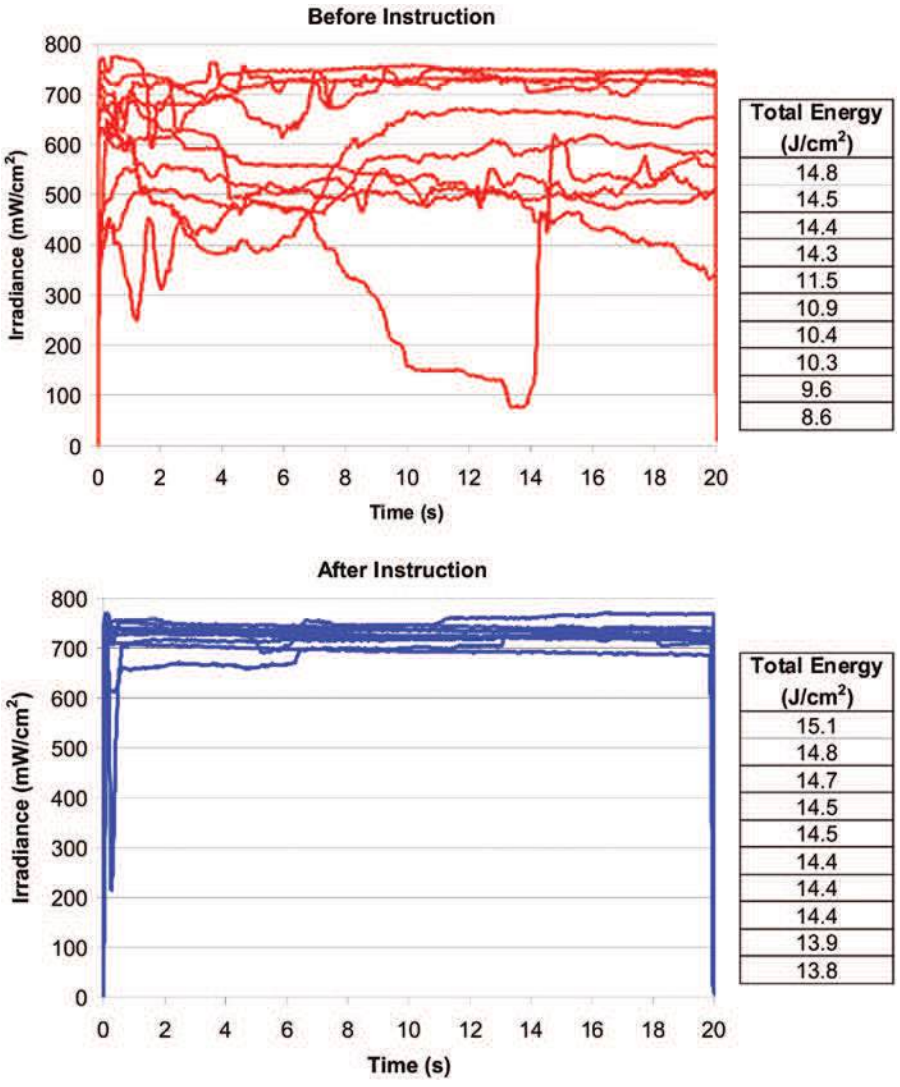
Much of the research on dental resins has been conducted in a laboratory setting, using a curing light held in a fixed position that is very close to the resin surface. Although this method makes the results more consistent, this is not a clinically relevant condition. When light curing a restoration in the mouth both the dentist and the patient often move, and access to the posterior teeth is often limited. Unless the operator is very careful, the position of the light tip can drift away from the restoration, or unwanted shadows may be produced that can have an adverse effect on the amount of energy delivered to the restoration and the subsequent level of resin polymerization (Fig. 6.51).[129,130]

It is known that the light-delivery technique used by the operator can have a significant effect on the radiant exposure delivered to the restorative material, especially when it is common practice not to watch the position of the curing light tip over the restoration when light curing. Not only can this practice negatively affect the amount of energy delivered to the restoration, it can also result in a tissue burn.[6,114,130-133] The operator variability in how much light is delivered can be reduced and the radiant exposure delivered to restorations increased if the operator has been trained using a device that shows, in real time, his or her ability to effectively deliver light energy to a simulated restoration. Such a clinical training simulator (MARC-Patient Simulator, BlueLight Analytics, Halifax, Canada) is seen in Fig. 6.52.

Individualized hands-on training on how to perform light curing of a restoration includes learning how to correctly position the patient to improve access and how to position the light guide throughout the light curing process.[129,131,133-135] By providing immediate feedback to the operator on how much irradiance and energy is being delivered, together with instructor coaching on how to avoid mistakes, the MARC-Patient Simulator has been shown to be an effective method to teach how to light cure a restoration. Examples of the real-time irradiance delivered to a specific restorative location by several operators during a 10-second exposure



• **Fig. 6.52** Light curing a simulated restoration using the MARC-Patient Simulator (BlueLight Analytics, Halifax, Canada).



• **Fig. 6.53** Before and after images of irradiance levels delivered by several operators to a specific intraoral location during a 10-second light exposure, as monitored using the MARC-PS patient simulator. The after image shows great improvement in how consistently the light was delivered throughout the exposures. The operators also delivered more total energy after instruction.

and monitored using the MARC-PS patient simulator are seen in Fig. 6.53. The after-image shows a great improvement in the consistency in the irradiance that was delivered throughout the exposure. The operators also delivered a greater radiant exposure (amount of energy in J/cm^2) after instruction, thereby increasing the potential for achieving the intended properties and the promising *in vitro* results for the restorative material they are using.

General Recommendations When Using a Curing Light

As a general recommendation, users should be aware of the radiant power output, active emitting tip dimensions, irradiance, effect of distance on irradiance, and emission spectrum from the curing light they are using. Their light should be tested regularly to ensure that it is functioning according to the manufacturer's specifications and that the output has not changed significantly since purchase. When light curing, clinicians should deliver sufficient radiant exposure at the correct wavelengths of light required by the photoinitiator used in the resin composite they are using. To maximize the radiant exposure delivered, the light tip should be held both close to and perpendicular to the surface of the restoration. Operators should directly observe what they are doing throughout the entire light exposure, and to do so safely, they must also use appropriate eye protection. Precautions should be taken to avoid causing an unacceptable temperature increase in the tooth and the adjacent soft tissues.[41,102,136]

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